

INSTITUTIONAL BIDDING IN IPO ALLOCATION: EVIDENCE FROM CHINA

Xiaping Cao

Lingnan College, Sun Yat-Sen University
&
Singapore Management University

Tiecheng Leng

Lingnan College, Sun Yat-Sen University

Bo Liu

University of Electronic Science and Technology of China

William Megginson

University of Oklahoma
King Fahd University of Petroleum & Minerals

Current draft: April 19, 2016

Abstract

Using a proprietary database of 123,819 bids by institutional investors for share allocations in 783 Chinese IPOs, we document that bidding dispersion significantly predicts share offer pricing. IPOs with higher levels of bid dispersion in the institutional tranche auctions experience greater first-day return than other IPOs by discounting the offer price as a compensation for investors' bearing valuation uncertainty and estimation risk. Our results hold after controlling for potential endogeneity and using alternative dispersion measures. Bid dispersion further negatively predicts both one-year operating performance post-IPO and three-month stock performance.

Keywords: Dispersion, Divergence of opinion, Institutional investors, Initial public offering, IPO underpricing, China, Auction

JEL classification: G20, G24

Institutional Bidding in IPO Allocation: Evidence from China*

Abstract

Using a proprietary database of 123,819 bids by institutional investors for share allocations in 783 Chinese IPOs, we document that bidding dispersion significantly predicts share offer pricing. IPOs with higher levels of bid dispersion in the institutional tranche auctions experience greater first-day return than other IPOs by discounting the offer price as a compensation for investors' bearing valuation uncertainty and estimation risk. Our results hold after controlling for potential endogeneity and using alternative dispersion measures. Bid dispersion further negatively predicts both one-year operating performance post-IPO and three-month stock performance.

Keywords: Dispersion, Divergence of opinion, Institutional investors, Initial public offering, IPO underpricing, China, Auction

JEL classification: G20, G24

* We gratefully acknowledge valuable comments from Tor Bakke, Chitru Fernando, Xuechen Gao, Bing Han, Kathleen Hanley, Ji-Chai Lin, Hamed Mahmudi, N. R. Prabhala, Yiming Qian, Ann Sherman, Gary Tian, William Wilhelm, Honghai Yu, and seminar participants at Deakin University, Monash University, Singapore Management University, University of Oklahoma, Southwestern University of Finance and Economics, the 2015 China International Conference in Finance and the 2016 American Finance Association annual meeting.

Institutional Bidding in IPO Allocation: Evidence from China

Initial Public Offerings (IPOs) typically experience large positive first-day returns in all the world's major capital markets (Ritter and Welch, 2002; Ljungqvist and Wilhelm, 2003; Lowry, Officer, and Schwert, 2010; Choi, Lee, and Megginson, 2010; Jagannathan, Jirnyi and Sherman, 2015), and empirical evidence suggests that Chinese IPOs may show the highest levels of underpricing globally, and have done so since the first post-Revolution stock market opened in China in 1990. Since that date, the Chinese IPO market has experienced rapid development, with the total market value of listed Chinese companies growing from virtually zero to RMB 62.75 trillion (USD 10.27 trillion) in May 2015. For example, in 2010 alone the market witnessed 347 IPO listings raising almost RMB 490 billion (USD 73.92 billion) in the domestic A-share market, including the initial offering of the Agricultural Bank of China, the world's largest IPO to that date.¹ The Chinese IPO market has also led the world in both number and aggregate value of IPOs almost every year since 2009 (Megginson, 2015). Unsurprisingly, because of its rapid growth and peculiar valuation patterns, China's IPO market has attracted great interest from academics, practitioners and policy makers.

China's system for launching IPOs has two remarkable distinctions from the general U.S. book-building system that is described in Benveniste and Spindt (1989), Benveniste and Wilhelm (1990), Cornelli and Goldreich (2001), Ljungqvist, Jenkinson, and Wilhelm (2003), Sherman (2005), and Fernando, Gatchev, May, and Megginson (2015). First, rather than having discretion over IPO share allocation and (until 2009) pricing, underwriters in China are heavily supervised by, and subject to the approval system of, the China Securities Regulatory Commission (CSRC) that regularly issues guidelines governing the listing process that IPO firms, underwriters, and institutional bidders must follow.² Second, in an information acquisition model, the underwriter's control of allocations through a traditional book-building system serves as a mechanism to induce institutional investors to reveal private information about IPO firms (Benveniste and Spindt, 1989; Aggarwal, Prabhalla, and Puri, 2002; Cornelli and Goldreich,

¹ See <http://www.reuters.com/article/china-privateequity-idUSTOE70904E20110110>.

² This official guidance typically involves asking issuing companies to refrain from overpricing, raising "excess" capital, and selling shares currently held by existing stakeholders. Underwriters and institutional investors consider such guidance to be *de facto* interference that often suppresses the offering prices of new IPO stocks. China's IPO processes and evolving regulations are described in Chan, Wang, and Wei (2004), Chen, Firth, and Xu (2009), Li, Wang, Cheung, and Jiang (2011), Chen, Shi, and Xu (2014), and Li, Megginson, Shen, and Sun (2016).

2003; Sherman, 2005), allowing underwriters to price IPO stocks efficiently.³ In fact, China's IPO underwriting system is a hybrid auction consisting of two tranches: an *auction tranche*, a competitive price-setting tranche in which only institutional investors are allowed to bid, and a separate fixed price *public pool tranche* that allows investors, primarily retail investors, to place orders without specifying a price. The auction tranche, the focus of this paper, uses a dirty multi-unit uniform price auction approach to allocate IPO shares to institutional investors. It allows everyone to pay the same offer price that is determined by the institutional bids, but the final offer price can be set lower than the market clearing price. Underwriters and issuers in China have discretion in setting the offer price, but not in share allocation, which is determined mechanically.⁴ Critics of this allocation feature argue that institutional investors have little incentive to produce valuable private information, since they cannot be rewarded with preferential share allocations. Rather they strategically inflate their bids to ensure receiving an allocation. China's auction tranche is different from Taiwan's discriminatory IPO auction in which bidders can participate in both tranches (Chiang, Qian, Sherman, 2010).⁵ Schnitzlein, Shao and Sherman (2015) offer experimental and theoretical evidence that the auction method for IPOs may be improved through the use of hybrid auctions with separate retail tranches or 'public pools' which increase proceeds, lower price volatility, reduce pricing error and reduce the incentive for small bidders to free ride by submitting extremely high bids. Whether China's hybrid IPO auction can facilitate the informational content of institutional investors' auction bids remains unclear and needs to be scrutinized.

³ Alternatively, Sherman (1992) models costly information production as a driving force for underpricing. Chemmanur (1993) argues that IPO firm insiders underprice their shares to induce outsiders to produce costly information. Derrien (2005) suggests that initial IPO returns result from market mispricing of the offer or closing price, while Purnanandam and Swaminathan (2004) argue that U.S. IPOs are overvalued compared to their industry peers.

⁴ In a multi-unit uniform price auction, bidders can bid for multiple units of the shares, and when the auction is closed, all winning bidders pay the uniform offer price, independent of their bidding price, to get the share allocations. China followed a *pro-rata* system proportional to the bidding volume prior to November 5, 2010, and then switched to a lottery system. If the offer price can be set lower than the market clearing price, it is known as a "dirty" auction or as "leaving something on the table". In a discriminatory, or pay-what-you-bid, auction, "winning" bidders pay their bid prices. Jagannathan, Jirnyi and Sherman (2015) define the terminology for dirty auctions for IPOs, and China's auction method during our 2009-2012 study period rather closely matches the process that these authors argue is best at price discovery and optimal share allocation achievable without the allocation discretion allowed in U.S.-style book building.

⁵ CSRC requires that all institutional investors only participate in one tranche after June 2009, and urges institutional investors to play better roles in setting offer prices and to improve the IPO underwriting mechanism. As a matter of fact, institutional investors generally choose to participate in the auction tranche.

According to Ritter (2011) and Ritter and Welch (2002), in their comprehensive reviews of the IPO literature, IPO bidding and allocation is an underexplored area because of data unavailability. In this paper, we take advantage of a large proprietary database, obtained directly from the Shenzhen Stock Exchange. The dataset contains detailed information on institutional bids (bidding time, bidding price, bidding quantity, and bidding institutions and their locations) for each IPO in China over the July 2009-November 2012 period. We empirically examine the informational role of institutional investors in the IPO bidding and allocation processes. Distinct from several other studies which have gained (often confidential) access to institutional bidding data in China to study IPO pricing and allocation, our database contains 123,819 institutional bids with RMB 13.18 trillion (USD 2.03 trillion) bid value. Our dataset is by far the most comprehensive in terms of number and value of bids, depth and length of coverage, and level of detail regarding six different types of institutional investors submitting bids.⁶ Our focus is on the role of institutional investors in setting offer prices and influencing first day returns, which contrasts with Chemmanur, Hu, and Huang (2010), who study the informational role of institutional investors in post-IPO trading.

In particular, we focus on the dispersion in institutional investors' bidding prices, which is analogous to the dispersion in analysts' earnings forecasts measure used in Imhoff and Lobo (1992), Barron, Kim, Lim, and Stuerke (1998), Barron and Stuerke (1998), and Diether, Malloy, and Scherbina (2002). All these studies show that dispersion is an appealing proxy for investors' divergence of opinion, and this is the first research to use a similar measure derived from institutional bids in IPOs. Since the IPO auction tranche is open exclusively to institutional investors, we then use bid prices of institutional investors to construct several clean and *ex ante* measures of opinion dispersion for each IPO. Our measures--unlike those employed in existing studies such as Houge, Loughran, Suchanek, and Yan (2001), which are based on *ex post* IPO market trading data and thus are likely contaminated by a forward looking bias that cannot reflect *ex ante* opinion dispersion--provide unbiased estimates of institutional opinion dispersion.

⁶ IPO studies employing institutional and/or retail investor bids include Hanley and Wilhelm (1995) [US], Aggarwal, Prabhala, and Puri (2002) [US], Cornelli and Goldreich (2003) [Europe], Jenkinson and Jones (2004) [Europe], Chiang, Qian, and Sherman (2010) [Taiwan], and Neupane and Poshakwale (2012) [India]. Liu (2014) hand collects a small subset of Chinese public available bidding data covering only the period November 2010-October 2012 with limited information disclosure is required by CSRC after November 2010, but his study examines (and documents) performance persistence of institutional investors. Our proprietary database is directly obtained from the Shenzhen Stock Exchange and is much more comprehensive with richer information than existing literature.

Another difference is that we not only study post-issue pricing or return but also examine pre-IPO opinion dispersion impacts offer pricing.

Our main hypothesis is the *divergence of opinion discount hypothesis* which posits that IPOs with greater institutional opinion divergence have higher value uncertainty and risk for which IPO investors need to be compensated. The alternative hypothesis is the *divergence of opinion premium hypothesis* which states that divergence of opinion prior to the IPO implies a lower future return due to short sale constraints and optimistic investors over-bidding. Empirically, we show that divergence of opinion is positively correlated with first-day share turnover, and also find that divergence of opinion negatively predicts the offer price but is uncorrelated with the first-day closing price. This leads to a strong and robust direct relationship between opinion divergence among institutional bidders and IPO underpricing, indicating that institutional bid divergence positively affects Chinese IPO first-day returns not by overvaluing the first-day close price, but instead by discounting the offer price. These results support the predictions of the *divergence of opinion discount hypothesis* regarding IPO pricing. Institutional bidding dispersion is shown to be negatively related to three-month stock return. Furthermore, institutional bidding dispersion is negatively predictive of one year post-IPO operating performance. Not only do these measures of institutional bidding dispersion have predictive power for underpricing and post-IPO performance, but the economic channels of influence are also identified.

To better understand the economic mechanisms behind our main results, we conduct several cross-sectional tests to exploit settings where the positive effect of opinion divergence on IPO first-day returns is predicted to be pronounced. Prior literature documents that the expected underpricing of an IPO increases with the *ex ante* valuation uncertainty about its value (Beatty and Ritter, 1986; Megginson and Weiss, 1991). Following the literature, we use offer size, time lag between offering and listing, and underwriter reputation to proxy for the *ex ante* uncertainty about an IPO. The results are clear and consistent across various proxies: the relation between opinion divergence and IPO underpricing is much more pronounced when the issuing firms' *ex ante* uncertainty is high. Other cross sectional tests show institutional investor's type and geographical distance between bidders and IPO firms matter for IPO first day return in a fashion consistent with *ex ante* uncertainty and an informational role for institutions.

To further address potential endogeneity problems, we take advantage of two CSRC-implemented reforms of IPO auction bidding in 2010 and 2012, using these as natural experiments. According to the CSRC's official "*Guidelines on Deepening Reform of New Issue System*" (CSRC announcement [2010]26) and "*Guiding Opinions on Further Reforming the Issue System of New Shares*" (CSRC announcement [2012]10), these two regulatory changes are aimed partly at improving institutional investors' bidding behaviour and discouraging them from simply bidding a superficially high price for IPO shares in the allocation process without revealing private information. The new rules increased the potential cost and reduced potential speculative profit margin in the secondary market for unserious bids. We find that these two regulatory changes have indeed caused the institutional bidder to bid more cautiously and led to a significant drop in institutional bidding dispersion while considerably augmenting its effect on IPO first-day returns. This finding suggests that institutional investors have improved their information production in the post-change period, which has made their bids more informative.

Our paper makes a useful contribution to a large body of literature examining IPOs in general and a smaller corpus studying the Chinese capital market in particular. Within this latter stream, Banerjee, Dai, and Shrestha (2011) find that Chinese IPOs have high first-day returns of 57.14%, while Chi and Padgett (2005) document initial first-day returns averaging over 100%. Chan, Wang, and Wei (2004), Chen, Firth, and Kim (2004), Chen, Firth, and Xu (2009), Firth, Lin, and Zou (2010), and Chen, Shi, and Xu (2014) further show that such returns are influenced by institutional environment, government, and ownership. In the broader literature, Berkman, Col, and Fu (2010) associate IPOs with securities market regulation and investor protection, while Dorn (2009) uses when-issued trades of German IPOs to show the importance of retail investor sentiment. None of these studies, however, examines IPO share allocations or studies the impact of institutional investors' opinion divergence on post-IPO performance. Our research thus helps illuminate how institutional opinion dispersion impacts the IPO market in general.

The remainder of the paper is organized as follows: Section I develops our testable hypotheses and profiles the unique institutional setting of the IPO market in China, after which Section II describes the data and variables. Section III presents the findings on the relation between IPO underpricing and the dispersion measure. Section IV examines the predictive power of the dispersion measure on firms' post-IPO performance, and Section V concludes the paper.

II. Hypotheses Development and Institutional Setting

A. *Hypotheses Development*

While divergence of opinion among investors is generally believed to play an important role in asset pricing, both theoretical predictions and empirical implications of divergence of investors' opinions on asset prices remain conflicted. On one hand, Miller (1977) takes divergence of opinion as a measure of heterogeneous beliefs and argues that when investors disagree on value, and when short-sale constraints are binding during a firm's public trading phase, the most pessimistic investors are driven out of the market and the most optimistic investors set stock prices, thus an IPO is priced *at a premium*. If future pricing efficiency can be assumed, then divergence of opinion prior to the IPO implies a lower future return. We refer to this as the *divergence of opinion premium hypothesis*. Alternatively, divergence of opinion among investors prior to the IPO may reflect valuation uncertainty or estimation risk resulting from limited information (Barry and Brown, 1986). In this case, divergence of opinion should lead to a positive risk premium (Varian, 1985; Merton, 1987; Abel, 1989; Epstein and Wang, 1994). This is because an increase in valuation uncertainty contributes to raising the required rate of return on the stock by *discounting its current market price* (given that future pricing efficiency can be assumed), relative to its true value as a compensation for investors bearing the risk. We refer to this as the *divergence of opinion discount hypothesis*.

Note that both hypotheses directly link the divergence of opinion to the offer price, and further imply a direct relationship between divergence of opinion and future equity return when future pricing is assumed to be efficient. This is important for IPO underpricing--which is measured by first-day return and defined as the ratio of the closing price of the stock on its first trading day less the offer price, divided by its offer price--since the offer price and the first-day closing price could be jointly affected and determined by divergence of opinion. Understanding this can help us clearly identify the real economic mechanism behind the relation between divergence of opinion and IPO underpricing. Specifically, the offer price is set by a dirty multi-unit uniform price auction, where both the most optimistic and most pessimistic investors can freely and fully express their opinions by submitting extremely high or low prices, respectively, thus directly affecting the final offer price since all institutional bids are aggregated to determine the offer price. Therefore, during this bidding and allocation stage, the *divergence of opinion premium hypothesis* may not hold since the necessary and sufficient condition of only optimistic

investors determining the offer price cannot be satisfied. Actually, in the Chinese IPOs' setting of sealed bid and common value auctions with uncertainty, investors don't even know how many shares other bidders are demanding before submitting their own bids. Thus, the *divergence of opinion discount hypothesis* may dominate and govern the offer price setting in that valuation uncertainty or estimation risk would obviously matter in this stage, given no prior trading history and limited financial information about the completely new shares created in the IPO. We hypothesize that institutional bidding dispersion is negatively related to IPO offer price.

On the first trading day, the short sale constraints are explicitly binding, and it seems that first-day closing price would be set at a premium and the *divergence of opinion premium hypothesis* should dominate, since the valuation uncertainty or estimation risk may be resolved to some extent through trading. Nevertheless, there is also the possibility that China's special institutional background and investors' structure may prevent the divergence of opinion from impacting the first-day closing price. First, the allocated institutions are subject to a three-month lockup provision for all but the last five months of our study period (718 out of 783 IPOs in our sample period are affected), so that the allocated shares cannot be sold within three months post-IPO. Second, retail investors from the public pool tranche account for most of the trading volume on the first trading day, but these investors cannot obtain the institutional investors' bidding information from the auction tranche (373 out of 783 IPOs in our sample period are affected), and they might underreact to this information even after they obtain it in first-day trading.

More importantly, due to the scarcity and high average underpricing of China IPOs, retail investors are always sentiment driven and rush to trade the new shares with limited attention to a firm's fundamentals and institutional bidding information (Cai, He and Zhai, 2015). These tendencies will reduce the timeliness of the information contained in the institutional investors' divergence of opinion and preclude it from being effectively impounded into the market in such a short trading period. Therefore, the divergence of opinion premium hypothesis may not necessarily predict the first-day closing price. This logic leads us to hypothesize that institutional bidding dispersion will either be positively correlated or uncorrelated with (but not negatively correlated with) the IPO's first-day closing price. Combined with the first hypothesis, we predict that institutional bidding dispersion is positively related to IPO underpricing.

If the *divergence of opinion discount hypothesis* dominates IPO pricing, valuation uncertainty or estimation risk resulting from limited information should be gradually resolved

through post-IPO trading, through which new information will be incorporated into market prices, and the previously discounted price should rebound back to the true value and lead to a lower future return. We refer to this as the *uncertainty resolution* channel. Alternatively, if the *divergence of opinion premium hypothesis* dominates IPO pricing, as short sale restrictions ease and additional firm information becomes available, short-term overvaluation might be corrected and prices should approach fundamental value. We refer to this as the *overvaluation correction* channel. Therefore, no matter which channel works, we hypothesize that institutional bidding dispersion will be negatively related to the IPO firm's long-run stock performance.

Fundamentally, given the limited information nature of IPO firms, institutional bidders rely heavily on a firm's historic operating performance to form their bidding strategies in the auction tranche, and a large divergence of opinion may convey a large disagreement among institutional investors and be a negative predictor of the firm's future operating performance. As institutional investors are recognized to play important roles in corporate governance (Gillan and Starks, 2000), investment behaviour (Bushee, 1998), and corporate operating performance (Cornett, Marcus, Saunders, and Tehranian, 2007), significant disagreement among institutional investors would also reduce market participants' confidence in the firm and affect the company's financing capacity and investment efficiency, negatively impacting the firm's future operating performance. We thus hypothesize that institutional bidding dispersion will be negatively related to the IPO firm's long-run operating performance. We also follow Garfinkel and Sokobin (2006) and hypothesize that institutional bidding dispersion is positively related to IPO price volatility and first-day share turnover. To test these hypotheses, we construct several measures of each IPO's opinion divergence using institutional bidding price dispersion.

Pricing a non-public issue is difficult and costly. Underwriters offer their private assessment on a valuation report (premarket information collected during the due diligence process), which is made available exclusively to institutional investors who participate in the auction tranche bidding. This is the only information other than that created by institutional investors' own efforts available to guide the pricing decision, thus institutional investors rely heavily on this report, especially the underwriter's valuation range. This special feature allows us to construct a test to address the endogeneity concern that unobserved IPO firm qualities may be correlated with dispersion and IPO first-day return or post-IPO performance. We run instrumental variable (IV) regressions using the relative underwriter valuation range deflated by

the midpoint valuation price as our instrumental variable. The underlying rationale is that this measure is unrelated to first-day returns but related to opinion dispersion, meaning that the narrower the range, the smaller the dispersion. The findings from our baseline tests remain robust in the IV regressions. The results also remain robust for alternative dispersion measures.

B. *Institutional Features of Chinese IPO Auctions*

The Chinese IPO market is notorious for high first-day returns that precede poor post-IPO performance, such as the average first-day returns of 247% or 145% in A shares traded on, respectively, the Shanghai and Shenzhen Stock Exchanges (Yu and Tse, 2006). Many blame these high first-day returns on speculation or noise trades by retail investors. Useful overviews of IPO underpricing in general and the development of the Chinese IPO market in particular are presented in Ritter (2011) and Ljungqvist (2004), respectively. We investigate these issues using a proprietary IPO dataset that is more detailed and comprehensive than those used by Cornelli and Goldreich (2001, 2003) and Jenkinson and Jones (2004) to examine actual orders and allocations in the European book building process. Unlike those studies, we concentrate on opinion divergence among institutional investors in a dirty multi-unit uniform price auction. Our proprietary sample of institutional bids includes all IPO companies that have gone public from July 2009 to November 2012 on the Shenzhen ChinNext Board and Shenzhen SME Board, both of which were established to allow small or growth firms to list shares and raise funds.

The IPO process in China has always been regulated by the CSRC, which has implemented many reforms to improve its approval-based system. For example, before June 2009, the CSRC implicitly imposed a price-earnings ratio cap of 30 for IPOs, meaning that the offering price could not be set over 30 times the firm's earnings per share. This limit obviously made IPO pricing highly inefficient. On June 11, 2009, the CSRC implemented its landmark "*Guiding Opinions on Further Reforming and Improving the Issuance System of New Shares*" (CSRC announcement [2009]13), which removed the implicit restrictions on the price-earnings ratio. This major reform marks the beginning of our sample period, during which two other important regulatory reforms were implemented in 2010 and 2012 for IPO pricing.

The typical IPO hybrid auction process in China during our sample period operates in auction tranche and public pool tranche (also called fixed-price public offer tranche) sequentially. Only institutional investors, and a very few experienced large individual investors recommended by the underwriter after the 2012 reform, are allowed to bid in the auction tranche, while the

public pool tranche mainly targets retail investors. There are usually two days between the tranches. In the auction tranche, an IPO firm chooses an investment bank as the lead underwriter, which is responsible for pricing, selling, and organizing the new issue. The CSRC has adopted a price inquiry system to implement the auction tranche bidding. The lead underwriter then invites all institutional investors in the recordation lists maintained by the SAC (Securities Association of China) to bid.⁷ The lead underwriters can also recommend institutional investors to bid in the IPO auction after November 5, 2010 and can recommend well qualified individual investors to bid in the IPO auction process after May 25, 2012. However, the size of these two types of bidders is trivial relative to other types of institutional investors.

The lead underwriters solicit sealed bids specifying prices and quantities from participating institutional investors over a certain period of time, typically two to three business days. Each institutional investor can submit at most three bids, which carry information on the number of shares to purchase and the price they are willing to pay. To encourage institutional bidders to bid more cautiously, the lead underwriter also provides bidders with its IPO valuation range and a detailed IPO valuation report for reference.⁸ The Shenzhen Stock Exchange also requires that no submitted bids can be cancelled in the bidding process. The underwriter monitors the dynamics of all the bids, including bidder identity, bid time, bid price, and bid quantity, but bidders themselves cannot see others' bids. The lead underwriter decides on a uniform offering price after collecting all the bids from the auction, and then reports the offering price to the CSRC for inspection.

Second, unlike most countries that have used strict sealed-bid procedures where no one knew the bids until the auction had closed, Chinese IPO underwriters can observe the full dynamics of all the bids (bidder identity, bidding time, bidding price, and bidding quantity) submitted in the IPO auction. This is another interesting and unique feature, which may create information leakage and sharing.⁹ For example, Jay Ritter's 2014 *Forbes* article reveals how underwriters leaked information to their most valuable clients, which affected customers'

⁷ Only one IPO has two lead underwriters; the other 782 IPOs in our sample have one lead underwriter each.

⁸ The complete IPO valuation report is only provided to the institutional investors bidding in auction tranche.

⁹ For WR Hambrecht OpenIPOs in the US (20 of the 23 IPO auctions), the prospectuses have frequently reserved the right to communicate "general auction trends" to institutional investors. Institutional investors have sometimes gotten calls from the underwriter suggesting that they change their bids based on what everyone else is bidding.

bidding strategies and share allocations in Google's 2004 IPO auction.¹⁰ The information leakage may come from an underwriter's rent-seeking behavior due to the close business relation with highly valuable customers (Ritter, 2011; Ritter and Zhang, 2007; Nanda, Wang, and Zheng, 2007; Gasper, Massa, and Matos, 2006; Reuter, 2006; Goldstein, Irvine, and Puckett, 2011). The bidders may wait until later stages of the auction process, when underwriters gather enough accurate information and pass the tips to them. If the bidder can receive private and accurate information about "general bidding trends" from underwriters, they can bid at lower but more accurate prices to obtain a higher winning probability at a lower cost. The potential for information leakage and sharing is obviously of great economic importance for the information content of bids.

Due to the extremely high excess demand for new shares in China, if the offer price was set at the market clearing price, it would be very high. Although a high offer price would allow the issuing firm to raise more proceeds and help the underwriter to obtain more underwriting commission, it would also shrink the profit margin of participating institutional bidders and secondary retail investors, thus discouraging their participation in future auctions. Additionally, to control the excess-proceeds problem, and "realize the balanced, coordinated and sound development of both the primary and the secondary markets, and tangibly protect the lawful rights and interests of investors", the CSRC tries to keep the offer price at a reasonable level. Consequently an excessively high offer price will be rejected by CSRC.¹¹ Thus, an underwriter's offer price setting in the auction tranche is not mechanically based on all the institutional investors' bids, but also incorporates discretion in the sense of trading off and balancing all the stakeholders' interests. All the bids above the final offer price are qualified for allocation. As previously emphasized, because the allocation rule is set by the CSRC, underwriters in Chinese IPOs, unlike those in U.S. IPOs, have no discretionary power in IPO share allocation. Instead, allocations follow a *pro-rata* system proportional to the bidding volume prior to November 5, 2010, after which allocations switched to a lottery system. The public pool tranche starts after the offer price is set, with investors subscribing at the fixed offer price and with allocation determined by balloting.

C. Two Major Reforms of Chinese IPO Auctions in 2010 and 2012

¹⁰ See <http://www.forbes.com/sites/jayritter/2014/08/07/googles-ipo-10-years-later/#23f4226d70f9>.

¹¹ See "Guiding Opinions on Further Reforming the Issue System of New Shares" (CSRC announcement [2012]10).

The CSRC's official files of "*Guidelines on Deepening Reform of New Issue System*" (CSRC announcement [2010]26) and "*Guiding Opinions on Further Reforming the Issue System of New Shares*" (CSRC announcement [2012]10) propose two batches of regulatory changes and clearly state that the targets of these two reforms are to "...continue to perfect the restriction mechanism on quotes and shares placement...enhance the responsibilities of price setters, and promote more cautious and truthful quotes..." and "...strengthening regulation of issue prices to urge issuers and all participants to fulfil their duties..."¹² Specifically, the major regulatory changes in 2010 reforms are as follows:

(1) *Set minimum bid size and change the allocation rule from pro-rata basis to lottery basis.*

Prior to November 5, 2010, IPO share allocation in China followed a *pro rata* system in which allocations were proportional to bidding volume.¹³ After this date, the allocation rule switched to a lottery system. In a pro-rata system, everyone's allocation is scaled down to a very small level, which obviously breeds unserious bids. Setting a minimum bid size to raise the participation cost and adopting a lottery system can avoid very small allocations, while simultaneously adding risk through wider variation in potential allocations and also wider variation in potential gains and losses for those allocated shares. This essentially increases the bidders' responsibilities and obligations for their own bidding. In anticipation of this, rational bidders will tend to engage in more information gathering and production and bid more cautiously. Actually, the underwriters always explicitly note that they intend to set a minimum bid size in order to make bidders bid more prudently during road shows.

(2) *Mandatory disclosure of institutional bids in auction tranche.*

The CSRC requires that issuers and underwriters disclose the institutional bids and allocation details publicly after the auction process is finished. The introduction of social monitoring on the bidders' behavior can make bidders act more prudently due to their reputation concerns.

¹² See also Page 40 of the 2010 CSRC Annual Report.

¹³ The qualified bids that institutional investors submit above the final offer price enter a lottery system through which shares are allocated by underwriters via random drawing. In this lottery, shares are allocated through a formula that multiplies the total bid volume by a lottery ratio, with the ratio indicating the percentage of subscribed capital that will eventually obtain a share. In addition, the CSRC allows for no discrimination against any investors (e.g., the size of bids and identity of bidders) in the auction tranche lottery process. Specifically, the CSRC requires that the issuing firms and their lead underwriters should notarize the lottery processes to ensure that share allocation is on a fair basis, which alleviates the concern about any potential selection bias of the successful bidders.

- (3) *Expand the scope of institutions participating in price inquiries and allow the lead underwriter to recommend a given number of long standing, qualified institutional investors with relatively high price setting capabilities for participating in auction tranche bidding and allocation.*

Including experienced institutional investors will increase the proportion of cautious bidders and strengthen the competition among the bidders in information production and auction bidding.

The major regulatory changes in the 2012 reforms are as follows:

- (1) *Raise the auction tranche quota to at least 50% of total shares.*

Raising the relative size of the auction tranche will increase the bidders' participation and price sensitivity to avoid unreasonable bids.

- (2) *Remove the three-month lockup period provision imposed on institutional bidders, and continue to improve regulatory measures for speculation in new shares to maintain the normal trading order of new shares.*

Excess demand and high speculative profit margins in the secondary market are what unserious bidders in the primary market rely upon. Increasing the quantity of tradable shares of new listed companies to effectively ease the share supply shortage and curb speculation in new shares will reduce expected secondary market profits, which will in turn make the bidders bid more cautiously.

- (3) *Further enlarge the scope of auction tranche participants.*

The lead underwriter may independently recommend five to ten individual investors with relatively rich investment experience to participate in the auction tranche. This will increase the proportion of cautious bidders and strengthen the competition among the bidders in information production and auction bidding.

- (4) *Strictly implement internal control rules and investment management business rules and further improve professionalism and standardization of pricing, and strengthen the regulation of the price inquiry and pricing processes and the punishment regarding road shows and unserious bids.*

Taken together, we expect that the 2010 and 2012 reforms will encourage institutional bidders to bid more prudently, improving information production. These discrete and substantial regulatory changes constitute a near perfect natural experiment, which we exploit to examine the

information content of the institutional bids and how information is incorporated into asset pricing.

II. Data and Variables

Our proprietary bidding and allocation database covers IPOs from the Shenzhen Stock Exchange (SZSE), one of the two major stock exchanges in mainland China. Our primary sample comprises the entire population of 783 Chinese firms listed on either the Shenzhen SME (Small and Medium-sized Enterprises) Board (428 listings) or the ChiNext Board (355 listings) between July 2009 and November 2012, which account for 88.5% of all IPOs listed in China during this period.¹⁴ Our proprietary database contains 123,819 bids with RMB 13.18 trillion (USD 2.03 trillion) bid value by institutional investors for share allocations in 783 Chinese IPOs, which implies RMB 16.83 billion involved in one IPO on average. This demonstrates that the IPO auction tranche is very active, competitive, and economically important.

This time period is selected for two reasons: First, it takes full advantage of China's first market-oriented new share issuance reform. Second, during this time window, the other key regulations on IPO pricing were kept relatively stable, with all IPOs using a dirty multi-unit uniform price auction.¹⁵ The cut-off date of November 2012 takes into account the CSRC suspending the Chinese IPO market in November 2012 for regulatory transition. Although it reopened in January 2014, the IPO allocation rules subsequently varied too sharply and frequently to be suitable for empirical research. Each order book contains detailed IPO bidding information, including number of bidders; bidder name, type (e.g., domestic brokerage firms, fund management firms, financial firms), and geographic location; date and time of bid submission; and bidding price and quantity. It also lists the final offer price and the number of actual shares allocated to each institutional bidder after the bidding period closes. We extract the lead underwriter's IPO valuation range data from his private valuation report in which the underwriter's pricing method, logic and evidence are thoroughly illustrated. This valuation report

¹⁴ There are only 102 IPOs on the Shanghai Stock Exchange from July 2009 to November 2012, and Shanghai Stock Exchange adopts a different two-step IPO allocation system and is not subject to allocation method change from pro-rata basis to lottery basis. Thus, the large sample of 783 IPOs with richer regulatory changes makes the Shenzhen stock market a more appropriate representation of China's first market-orientated IPO reform starting from June 2009 and a better setting to examine the information content of institutional bids.

¹⁵ A partial auction mechanism was initially introduced by the China Securities Regulatory Commission (CSRC) in January 2005 and then subsequently altered and enforced in June 2009.

is only provided to institutional bidders and is not posted publicly. In practice, due to the limited information for new IPO firms, bidders rely heavily on this valuation report to form their bidding strategies.

We obtain the IPO firm financials, issue-specific characteristics, underwriter information, and stock market conditions directly from the Chinese Stock Market Accounting Research (CSMAR) and Wind databases. Whenever information is missing or incomplete in either database, we manually search prospectus for these data. To further identify the source of information content of institutional bids, we use geographic coordinates to calculate direct and indirect flight distances and Euclidian distances between the issuing firm and each institutional bidder in every IPO.

A. *Dispersion Measures*

In the Sherman (2005) model of costly information production and endogenous entry, bidders have the choice of whether to purchase a noisy signal as well as whether or not to bid. The model implies that a lower cost of information would attract more bidders to become informed, and lead to a tighter spread between signals, meaning a tighter dispersion of bids and predicting a more informative and accurate offer price. To explore the information content of institutional bids--distinct from the general bid clustering and “elasticity” analysis by Kandel, Sarig and Wohl (1999) and Degeorge, Derrien and Womack (2010), which may suffer the confusion of strategic or collusive bid-shaving with bid-shaving for the winner’s curse--we use dispersion of bids as an information aggregation measure. Bids reflect all available information that the bidder has, so the dispersion of bids is a sufficient statistic for their information sets, and hence is informative. A more cautious bid is generally made by exploiting richer available information, and thus should be more informative.

Analyst forecast dispersion is amply documented in the finance and accounting literature as a strong predictor of future stock returns and as a good proxy for differences in opinion among investors. For example, Diether, Malloy, and Scherbina (2002) show the dispersion in analysts’ earnings forecasts to be very useful in formulating profitable trading strategies because stocks with high analyst forecast dispersion are associated with a future return discount, especially in small firms. They attribute this negative relation to market frictions resulting from a lack of consensus among investors that limits the short sales of pessimistic investors and temporarily drives stocks into overpricing. Johnson (2004), on the other hand, develops a simple rational

asset pricing model in which dispersion proxies for the unpriced information risk arising from unobservable asset valuations, and argues that higher estimation risk leads to higher stock price and subsequently lower expected returns for risky, levered firms.

It should also be emphasized that institutional investors are sophisticated and often have an informational advantage over individual investors around various corporate events. Indeed, the extant literature documents that institutional investors outperform individuals, either because institutions have some unique private information that individuals do not have or because they can better interpret readily available public information (Aggarwal, Prabhala, and Puri, 2002; Chiang, Qian, Sherman, 2010; Nupane and Poshakwale (2012); Liu, 2014; Jagannathan, Jirnyi and Sherman, 2015). For example, Field and Lowry (2009), in their analysis of institutional holdings in newly public firms, demonstrate that firms attracting the highest levels of institutional investment significantly outperform those with the lowest levels. They attribute institutional investors' superior returns to their ability to better interpret public data. By similarly focusing on the role of institutional investors in equity issuances, Chemmanur, He, and Hu (2009) and Chemmanur, Hu, and Huang (2010) show that institutions systematically possess private information that enables them to realize superior returns over individual investors in both IPOs and SEOs.

Given the popularity of analyst forecast dispersion as a proxy for risk and uncertainty in the literature, we first use institutional investors' bidding price information to construct a simple measure of their heterogeneous beliefs in the pre-IPO market and then relate this variable to subsequent IPO underpricing. Similarly to what Diether, Malloy, and Scherbina (2002) do for earnings forecast dispersion, we measure the degree of heterogeneous beliefs among institutional investors in the auction tranche as their bidding price dispersion, defined as the ratio of the cross-sectional standard deviation of institutional investors' bidding prices in the auction bidding process scaled by the mean bidding price:

$$STD = \frac{\sqrt{\frac{\sum_{i=1}^N (p_i - \bar{p})^2}{N-1}}}{\bar{p}}$$

Where p_i is the bidding price from institutional bidder i and $\bar{p}$ is the average bidding price among all the N institutional bidders in an IPO. The dispersion in investors' bidding prices is a forward-looking measure that takes into account institutional bidders' heterogeneous beliefs about the issuing firm's quality and future profitability. We also construct two alternative dispersion

measures, *MAD*, the mean absolute deviation of the bidding price surrounding the mean bidding price scaled by the mean bidding price, and *STD weighted*, the bidding price dispersion among institutional investors weighted by their bidding volume.

B. Summary Statistics

Following the existing IPO literature, we define IPO first-day underpricing as the ratio of the stock's closing price on its first trading day less the offer price to its offer price. Panel A of Table I reports the summary statistics of our primary analytic variables, all of which are defined in Appendix A. As the panel shows, the average IPO first-day return for the entire sample of IPO firms is 36.84%, with a median value of 27.19%. Since the China Company Act of 1993 prohibits secondary share offerings in IPOs, all these issues are sales of newly issued, primary shares. In terms of general characteristics, our sample firms are on average 8.33 years old at the time of the IPO, the average time gap between offering and listing is 16.6 days, and the average issue size is RMB 1,061 million (USD 170 million using the 2010 exchange rate), with a median of RMB 625 million (USD 101 million). On average, there are 71 qualified institutional bidders participating in the auction tranche share allocation process for each IPO, with an average oversubscription rate of around 107 times the auction tranche. About 50% of our sample firms receive financing from either venture capital or private equity before going public. The vast majority of these offerings are from private companies; only 59 of the 783 issuing firms are state-owned enterprises. As noted previously, there are 373 IPOs offered before the CSRC's second round of IPO reform in 2010, another 410 offered in the 2010-2012 post-reform period, and a final 65 IPOs offered after the CSRC's third round of IPO reform in 2012.

****** Insert Table I about here ******

Panel B of Table I, which reports the Spearman correlation matrix for the primary analytic variables, shows an insignificant correlation coefficient of -0.01 between IPO underpricing and the *STD* dispersion measure. It also indicates that underpricing is significantly higher for IPOs with a smaller offer size and a longer time gap between offering and listing, as well as for IPOs that attract more institutional bidders.

We follow Degeorge, Derrien and Womack (2010) and define a free-riding bid as one where the bid price exceeds the top of the underwriter's IPO valuation range by more than 20%. The free-riding level for a given IPO is defined as the ratio of the number of free-riding bids divided by its total number of bids. Panel C of Table I reports the changing dynamics of the free-

riding level in the full sample and across CSRC's two major IPO reforms. Panel C of Table I shows that the average free-riding level is 8.54% over the entire study period. Before the 2010 IPO reform, the free-riding level is only 15.14%, and this falls sharply to 2.54% after the 2010 IPO reform. In the post-2010 reform period but before the 2012 IPO reform, the free-riding level is only 2.97%, and declines dramatically to 0.29% after the 2012 reform. This indicates that the free-riding level significantly falls after the CSRC's second round IPO reform, and further reduces after the third round IPO reform, suggesting that the two IPO regulatory changes indeed discourage free-riding behavior, and less free-riding may imply higher information content of the institutional bids.

Graphing the number of IPOs and average IPO underpricing on a quarterly basis, however, reveals that both measures fluctuate significantly over time, as shown in Figure 1. The number of IPOs ranges from 83 in 2010Q2 to only four in 2012Q4. IPO underpricing also fluctuates significantly. Toward the end of our sample period, in 2012Q4, the quarterly number of IPOs drops dramatically while IPO underpricing spikes to an extremely high level. This somewhat surprising trend is probably due to the CSRC's temporary ban on IPOs in mainland China's IPO market after November 2012, just before which investor enthusiasm for the relative scarce new shares in the secondary market pushed first-day underpricing to unreasonable levels.

****** Insert Figure 1 about here ******

III. Dispersion and IPO Underpricing

The first of our multivariate analyses of the explanatory power of institutional dispersion for IPO underpricing is a baseline regression using ordinary least squares (OLS). To address endogeneity concerns, we run an instrumental variable estimation in which the ratio of the lead underwriter's IPO valuation band to the midpoint of the valuation band is the instrument. Using these estimations, we show that our findings are robust to two alternative measures of bidder dispersion in the pre-IPO market. We also run subsample regressions based on issuers' pre-IPO information environment to identify how these amplify the effect of institutional bidders' heterogeneous beliefs on IPO underpricing. Finally, we use the natural experiments of the November 2010 and May 2012 IPO reforms to test the robustness of our findings.

A. Baseline Results

As in the existing IPO literature, we control for a rich set of firm and issue characteristics that may affect IPO offer price, first-day closing price and underpricing. For example, like Ritter (1984), Beatty and Ritter (1986), Carter and Manaster (1990), and Chiang, Qian, and Sherman (2010), we consider firm age, offer size, underwriter reputation, and the time gap between offering and listing as measures of the issuing firm's ex-ante uncertainty. To control for overall market conditions at the time of the IPO, we follow McGuinness (1992) by also including Shenzhen A-share market returns over one month prior to the listing date. We capture the profitability of the issuing firm by including return on equity (ROE) for the last fiscal year preceding the IPO, and use the share oversubscription rate and number of institutional participants in the auction tranche to control for the aggregate premarket demand for the issue. Because both venture capital (VC) and private equity (PE) sponsors are subject to reputation concerns, we include an indicator variable to flag issuers backed by either VC (Megginson and Weiss, 1991) or PE. We also use an indicator variable to differentiate IPOs listed on the ChiNext Board from those listed on the SME Board and include two separate indicator dummy variables to capture the effects of the two major IPO regulatory reforms introduced by the CSRC during our sample period. In all regression specifications, we include industry (based on CSRC classifications) and year fixed effects to account for potential industry and time trends. Following Petersen (2009), we compute heteroskedasticity-adjusted standard errors clustered at the industry level.

Our baseline OLS regression is specified as follows:

$$\begin{aligned}
 \text{IPO Initial Return} = & \alpha + \beta_1 * \text{Dispersion} + \beta_2 \text{Firm age} + \beta_3 \text{Offer size} \\
 & + \beta_4 \text{Time gap} + \beta_5 \text{ROE} + \beta_6 \text{Index return} + \beta_7 \text{Oversubscription} \\
 & + \beta_8 \text{Log \# Institutions} + \beta_9 \text{VC/PE dummy} + \beta_{10} \text{ReguChange1} \\
 & + \beta_{11} \text{ReguChange2} + \beta_{12} \text{ChiNext dummy} + \beta_{13} \text{Lead reputation} + \varepsilon \quad (1)
 \end{aligned}$$

The baseline regression results regarding the impact of opinion dispersion on offer price, first-day closing price and underpricing are reported in Table II. These results show a positive relationship between bidding price dispersion and IPO first-day returns that is significant at the 1% level. We further notice that the bidding dispersion is significantly negatively related to offer price but uncorrelated with first-day closing price. This implies that bidding dispersion affects underpricing by discounting the offer price through the channel of compensating investors for bearing valuation uncertainty and estimation risk resulting from the limited information nature of

IPO firms. In terms of economic magnitude, the coefficient estimates in column (3) suggest that a one standard deviation increase in the bidding price dispersion measure translates into a 5.28 percentage point ($=1.57 \times 0.0336$) increase in the IPO first-day return. This outcome represents an economically significant 14.33% increase in first-day underpricing relative to the average first-day underpricing of 36.84% in our full sample.

****** Insert Table II about here ******

Since the IPOs offered in the post-2010 IPO reform period have, on average, significantly higher ROEs (0.256 versus 0.216), a higher ratio of VC/PE backed firms (0.529 versus 0.467), a higher ratio of ChiNext firms (0.522 vs. 0.378), and higher underwriter reputation (3.731 versus 3.295), they are expected to obtain higher valuation than IPOs in the pre-2010 IPO reform period. The significant positive relation between IPO underpricing and the *ReguChange1* indicates that although the CSRC's 2010 IPO reform does discourage the institutional bidders' free riding of bidding at very high prices (quasi-market orders) and makes investors bid more cautiously, market valuation is relatively higher after 2010 relative to pre-2010 IPO reform period, leading to a higher offer price and first-day closing price. Furthermore, an even higher first-day closing price in secondary market trading after the 2010 reforms yields higher average underpricing.

Among the other control variables, smaller issues and those with a longer time gaps between offering and listing are associated with greater underpricing, as is a better overall pre-IPO stock market performance. Consistent with Cornelli and Goldreich (2003), issues that can attract higher premarket demand, as proxied by a higher oversubscription rate and more institutional participants, tend to have higher underpricing. On the other hand, issues listed on the ChiNext Board are significantly more likely to experience lower underpricing. We also find that VC or PE backed IPOs do not experience significantly greater underpricing. Overall, our baseline results suggest that IPO-related opinion divergence among institutional investors in the pre-IPO market has strong predictive power for IPO underpricing.

B. Instrumental Variable Approach

The positive relation reported in the baseline regressions may reflect one or both of two competing explanations: opinion divergence among bidders may in fact be based on some private information about the IPOs or it may be driven purely by the endogenous matching between institutional bidders and IPO firms. That is, if unobservable variables are simultaneously driving the relation between bidders' opinion divergence and IPO first-day

returns, the observed relation reflects the endogenous nature of the bidder's opinion divergence rather than any private information about first-day returns, thereby biasing the OLS estimates. We address this possible endogeneity between bidders' premarket opinion divergence and subsequent IPO underpricing by estimating a two-stage least square (2SLS) regression in which the ratio of the lead underwriter's IPO valuation band to the valuation band midpoint, $2 \times \left(\frac{\text{Prc}_H - \text{Prc}_L}{\text{Prc}_H + \text{Prc}_L} \right)$, instruments for this divergence, as measured by the bidding price dispersion *STD*. For the relative ratio measure to be considered a valid instrumental variable (IV), it must satisfy both relevance and exclusion conditions (Roberts and Whited, 2012). We choose this instrument because a tighter valuation band leaves less room for bidding price variation, resulting in relatively smaller bidding dispersion. Hence, holding everything else constant, our instrument should be positively correlated with the price dispersion and thus the relative ratio measure should satisfy the relevance condition as an instrument. On the other hand, because this relative ratio measure contains little direct information on price discovery, it is not immediately apparent how the ratio should drive the underpricing of a particular IPO in a way other than through its effect on bidding price dispersion. We thus feel fairly confident that our instrument satisfies the exclusion restriction as well.

****** Insert Table III about here ******

Panels A and B of Table III report the results of the first- and second-stage IV regressions, respectively. In the first-stage regression, the relative ratio of the lead underwriter's IPO valuation band to the valuation band midpoint is used as an IV for our dispersion measure *STD*. In the second-stage regression, we replace the dispersion measure *STD* with its predicted value from the first stage, the *Fitted STD*, and re-estimate the baseline regression. The first-stage results (Panel A) indicate that the proposed instrument is positively and significantly correlated with the bidding price dispersion *STD*, confirming that a tight valuation range helps reduce bidding price dispersion. The F-statistic is 49.80 and the R^2 of the regression is 24.6%, suggesting that our instrument satisfies the relevance condition and we do not have a weak instrument problem. The second-stage results (Panel B) then show that the instrumented bidding price dispersion-*Fitted STD*, also remains positive and significant at the 1% level, suggesting that IPOs with a higher level of premarket bidder opinion divergence subsequently experience greater underpricing. Moreover, the magnitudes of the key coefficients are also similar to those from the baseline OLS regressions. Collectively, our 2SLS analysis confirms that the positive relation

between bidder opinion divergence and IPO first-day return is unlikely to be biased by endogeneity.

C. *Alternative Dispersion Measures*

We further test the robustness of our main dispersion measure by constructing two alternative measures of IPO-related bidder opinion divergence. Consistent with prior accounting literature (Jacob, Lys, and Neale, 1999; Barniv, Myring, and Thomas, 2005), our first alternative dispersion measure *MAD* is the simple average of the cross-sectional unsigned mean absolute deviation of individual bidding price from the mean bidding price, scaled by the mean bidding price. Specifically, we calculate *MAD* using the following formula in which p_i is the bidding price from institutional investor i and $\bar{p}$ is the mean bidding price among all the N institutional investors participating in the auction tranche of an IPO:

$$MAD = \frac{\frac{1}{N} \sum_{i=1}^N |p_i - \bar{p}|}{\bar{p}}$$

Our second alternative measure, *STD weighted*, takes into account the importance of bidders' opinions about an IPO in the whole bidder group by weighting each institutional bid according to bidding quantity. Hence, *STD weighted*, formulated as shown below, gives larger bids greater weight relative to smaller bids:

$$STD \text{ weighted} = \frac{\sqrt{\frac{\sum_{i=1}^N w_i (p_i - \bar{p}_w)^2}{\frac{(N-1) \sum_{i=1}^N w_i}{N}}}}{\bar{p}_w}$$

where p_i is the bidding price from institutional investor i and $\bar{p}_w$ is the average bidding price weighted by corresponding bidding quantity among all the N institutional investors participating in the auction tranche of an IPO.

****** Insert Table IV about here ******

Table IV presents regression results using the two alternative dispersion measures, with specifications (1) and (3) LACKING industry or year fixed effects, but specifications (2) and (4) containing both. Here, using *MAD* and *STD weighted*, we obtain quantitatively and qualitatively similar outcomes to the baseline results shown in Table II. Across all regression specifications, both alternative dispersion measures are significantly and positively correlated with IPO first-day return at the 1% levels. In unreported analyses, we show that our main findings are robust to using alternative measures of opinion divergence among bidders and remain robust when we test

them using *price update*, defined as the percentage difference between the midpoint of the lead underwriter's IPO valuation range and the final offer price (Aggarwal, Prabhala, and Puri, 2002), with market-adjusted IPO first-day return as the dependent variable. They also hold when we exclude financial firms or the four IPOs with extremely high underpricing levels in the last quarter of our sample period, either separately or simultaneously.

There might also be potential concern for the strategic over-bidding of institutional investors in the auction tranche. As such, their opinion dispersion might be biased and not informative. However, the bidding and allocation mechanism in China helps mitigate such motives of over-bidding institutional investors. First, providing a high bid price can be costly for institutional investors, because the offer price is set based on aggregating all the bids. Over-bidding could systematically pull the offer price to a higher level so that all institutions have to buy the shares at relatively inflated offer prices and suffer high probability of the offer price breaking (the market price falling below the offer price) in the aftermarket, thus incurring large losses. Second, investors have to deposit enough cash to cover their bid value into a special escrow account when submitting bids, with the deposit frozen until the share allocation process in the auction tranche is completed. Because institutional investors must borrow or liquidate their current investment portfolios to raise funds to participate in the auction bidding, and because short-term Chinese interest rates often jump in advance of big IPOs, this implies a very large participation cost in the form of committed and frozen liquidity. Interestingly, the CSRC announced at year-end 2015 that this rule would be waived from 2016 onward, but in our sample this constraint is still binding. Finally, as a further robustness check, we calculate our opinion dispersion measure by removing the top 10% of bids with the highest bid price quotations and re-estimating the baseline regressions. We find that our main results remain qualitatively unchanged, thus ruling out the possibility that institutions' strategic over-bids drive the results.

D. Cross-Sectional Analysis

We conduct two additional sets of cross-sectional analyses: the first identifies the role of information asymmetry in the relation between opinion divergence among IPO bidders and subsequent IPO underpricing, and the second assesses the heterogeneity in the predictive power of opinion divergence across different bidder categories given the availability of information on institutional bidder types.

D.1. Impact of Ex-ante Uncertainty on the Dispersion-Underpricing Relation

Because of prior evidence that IPO underpricing increases with ex-ante uncertainty about issue value (Beatty and Ritter, 1986; Megginson and Weiss, 1991), we predict the effect of opinion dispersion on IPO first-day returns will vary across uncertainty levels. For example, Mok and Hui (1998) find that a lengthy time gap between the offering and listing of Chinese A-share IPOs increases *ex-ante* issuer uncertainty. To test this assumption, we partition the full sample into subgroups based on issuer *ex-ante* uncertainty as proxied by offer size, time gap between offering and listing, and underwriter reputation. We then re-estimate the baseline regression in Equation (1) for each subgroup. The odd (even) number columns in Table V report the outcomes for the subsamples with greater (smaller) ex-ante uncertainty. We note first that the forecasting power of our dispersion measure is significant and positive across all subsamples, indicating that our results are not subject to a sample selection bias. Consistent with our cross-sectional prediction, the relation between institutional dispersion and the level of IPO underpricing is more pronounced in smaller IPOs, IPOs with a shorter time gap between offering and listing, and IPOs managed by less reputable underwriters, computed for China using a Megginson and Weiss (1991) metric, similar to Chen, Shi, and Xu (2014). For example, in the subsample with a below median time gap (column 3), a one standard deviation increase in the dispersion measure translates into a 7.4% ($=2.273 \times 0.0326$) increase in IPO first-day return, whereas in the subsample with an above median time gap (column 4), the increase is only 1.79% ($=0.527 \times 0.0340$). Overall, these results indicate that ex-ante issue uncertainty amplifies the effect of bidder opinion divergence on expected IPO first-day returns.

****** Insert Table V about here ******

D.2. Heterogeneity of Investor Type

Because Ke and Ramalingegowda (2005) demonstrate that different types of institutional investors possess different private information about firms' future earnings and return, we classify each institutional investor according to the registration type recorded by the Securities Association of China (SAC). The resulting categories are as follows: domestic brokerage firm (BF), fund management firm (FM), financial firm (FF), trust company (TC), insurance company (IC), or qualified foreign institutional investor (QF). We then construct a dispersion measure for opinion divergence within each bidder category and explore the heterogeneity in this divergence's predictive power across categories. Because brokerage firms and fund management firms have their own equity research teams and expertise in generating IPO information, and also

bid more cautiously¹⁶, we expect divergence in BF to have the most information content and thus the strongest forecasting power in determining IPO underpricing, followed by divergence in FM.

Table VI reports the regression results for the separate bidder categories, with *STD_BF* representing the dispersion measure among domestic brokerage firms; *STD_FM*, fund management firms; *STD_FF*, financial firms; *STD_TC*, trust companies; *STD_IC*, insurance companies; and *STD_QF*, qualified foreign institutional investors. For all regression specifications, we use the same set of control variables and include industry and year fixed effects. Consistent with our expectations, we find that opinion divergence within the brokerage firm and fund management firm categories has the strongest predictive power for IPO underpricing. We find no strong relation, however, between IPO underpricing and opinion divergence within the remaining bidder categories (financial firms, trust companies, insurance companies, or qualified foreign institutional investors). Overall, our findings support the existence of a considerable amount of heterogeneity in the predictive power of opinion divergence across different bidder categories.

****** Insert Table VI about here ******

E. Robustness Test with the Natural Experiment of the Share Allocation Reform

The November 2010 and May 2012 reforms of IPO auction share allocations represent two exogenous shocks to institutional investors' bidding strategies. As we noted earlier, according to the CSRC's guideline, these two reforms were supposed to enhance the role of institutional investors in IPO price discovery by encouraging them to bid more cautiously and thus produce less biased bidding prices in the IPO auction process. The mandatory adoptions of these two regulatory changes thus serve as natural experiments that allow us to investigate the information role of institutional IPO investors. It is also important to understand the consequences of the reforms. To do so, we perform two sets of empirical analyses: the first determines whether and how these two reforms have affected bidding behaviors such as dispersion among the institutional bidders in auction bidding; the second assesses the impact of these exogenous regulatory shifts on the relation between the opinion divergence of institutional bidders and IPO underpricing.

¹⁶ Fund management firms usually have a serious and strict decision process on IPO shares bidding, such as establishing special committees to collectively (not individually) evaluate the IPO firm and decide the bidding strategy, and setting up the post-IPO performance evaluation mechanism on the fund manager's auction bids.

Especially regarding the 2012 IPO reform, we focus on a sample of 410 IPOs listed after the 2010 reform to ensure that no other major policy shifts can bias our analysis during this sample period. Out of the 410 IPOs, only institutional bidders in the last 65 IPOs are affected by the 2012 reform. To minimize concerns that some cross-sectional heterogeneity of IPO firms in our sample changes systematically with the 2012 reform, we construct a balanced sample, where each of the 65 IPOs after 2012 reform is matched with a comparable IPO from a pool of 345 IPOs before the 2012 reform but after the 2010 reform based on several key observable characteristics. To achieve a matched sample, we select matching IPOs that have the same industry classification and are listed on the same board of the sample IPOs, while minimizing the Mahalanobis distance metric based on the natural logarithm of firm age and ROE.¹⁷ The final sample consists of 65 matched IPOs before the 2012 reform and 65 IPOs after the 2012 reform for a total of 130 IPOs.

Table VII reports the multivariate regression results that use opinion divergence of institutional investors, *STD*, as the dependent variable. The explanatory variables of interest are the two regulatory change dummies: *ReguChange1* and *ReguChange2*. *ReguChange1* equals one if the IPO bidding date is after November 5, 2010, when the CSRC conducted the second round IPO reform, and zero before then. *ReguChange2* equals to one if the IPO listing date is after May 25, 2012, when the CSRC conducted the third round IPO reform, and zero before then. We find that both *ReguChange1* and *ReguChange2* are negatively and significantly related to the *STD* dispersion measure, suggesting that the CSRC's two reforms have been effective in lowering opinion dispersion among institutions, and institutional bidders bid more precisely. The effect is also sizable. For example, after the 2012 reform, bidders' opinion divergence is reduced by approximately 4.7% ($=0.0070/0.1486$), relative to the average dispersion measure of the 65 matched IPOs before the 2012 reform (but after 2010 reform).

****** Insert Table VI about here ******

We next investigate how these two reforms affect the relation between investor opinion divergence and IPO first-day returns. Specification (1) in Table VIII includes an interaction term $STD \times ReguChange1$ to gauge the interaction effect between the 2010 reform and the dispersion measure of IPO underpricing. The coefficient of the interaction term is positive and significant.

¹⁷ In alternative models, we match and balance the sample based on *coarsened exact matching* or *propensity score matching*, taking only the nearest neighbor of each sample IPO firm, and the results are qualitatively the same.

This finding indicates that the relation between opinion dispersion and IPO underpricing is more pronounced after the 2010 IPO reform. In specifications (2) and (3), we split the sample into two groups of pre-2010 reform and post-2010 reform and run the same regression separately on each. Comparing the outcomes for the two subgroups clearly shows that the 2010 IPO reform has amplified the effect of investor opinion divergence on IPO underpricing. For example, after the 2010 IPO reform (3), a one standard deviation increase in the dispersion measure translates into a 7.7% increase in IPO underpricing versus only a 3.2% increase before the change (2).

In addition, we also find that bidders' opinion divergence has a larger impact on IPO underpricing after the 2012 reform. For example, in the subsample of IPOs after the 2012 reform (specification 6), a one standard deviation increase in dispersion translates into a 22.01% ($=7.08 \times 0.031$) increase in IPO first-day returns, whereas in the subsample of matched IPOs before the 2012 reform (specification 5), the increase is only 7.32% ($=2.187 \times 0.033$). The difference in the coefficient estimates of the *ReguChange2* between the two subgroups is also significant, as indicated by the significant positive interaction term in specification (4).

Overall, the findings reveal that the positive relationship between opinion divergence and IPO underpricing is amplified by the 2010 and 2012 reforms. The empirical evidence from the regulatory changes as two natural experiments thus suggests that the relation between opinion dispersion and IPO first-day return is not caused by omitted variables or endogeneity problems.

**** Insert Table VIII about here ****

IV. Dispersion and Post-IPO Firm Performance

Having identified the relation between institutional bidding dispersion and IPO pricing efficiency, we now examine whether such dispersion is predictive of subsequent IPO stock returns and post-IPO operating performance.

A. *Dispersion and Post-IPO Stock Performance*

Our primary measures for post-issue firm stock price performance are one-, three-, and six-month post-IPO buy-and-hold returns (BHRs), which are calculated based on monthly stock returns beginning with the first month after the IPO listing date. Cumulative abnormal returns (CARs) are the cumulative difference between the monthly return of a particular IPO and the monthly return of the corresponding value-weighted market index. Because institutional bidders in the 718 IPOs listed before May 2012 are subject to a three-month lockup provision, however,

we expect that the overall predictive pattern of opinion divergence on stock performance will differ between bidders before and after this date. Specifically, the provision will prevent the opinions of the constrained bidders from being too quickly impounded into stock prices because they cannot trade shares obtained in auction tranche within the lockup period, meaning that such information can only be reflected in the stock price after the lockup provision expires. In contrast, the opinion divergence among bidders who are not constrained by the provision after May 2012 can be expected to predict one-month stock performance because their opinions can be fully revealed immediately after trading starts.

According to Panel A of Table IX, which presents the summary statistics for the post-issue one-, three-, and six-month firm stock performance measured as BHRs and CARs, respectively, newly listed firms on average exhibit negative BHRs over the six-month post-issue period. To assess the predictive power of premarket bidder opinion divergence for subsequent post-issue stock performance using full sample IPOs and IPOs with and without the three-month lockup provision, we use the baseline regression specification but include the corresponding holding period market index returns. We first note from Panel B of Table IX that the association between bidder opinion divergence and BHRs three months post IPO is negative and significant at the 1% level but becomes insignificant one month and six months post IPO, seemingly confirming our main argument.¹⁸

****** Insert Table IX about here ******

Since many IPOs in our sample are subject to the three-month lockup provision, we also run subsample regressions that explicitly separate out the effect of the lockup provision on the predictive pattern of bidder opinion divergence on post-issue stock performance. Consistent with our expectations, the estimates in columns (4) to (6) in Panel B of Table IX show that the opinions of bidders subject to the three-month lockup provision can only predict stock performance beginning three months post IPO, indicating that bidder opinion divergence contains private information about post-issue stock performance that can only be impounded into stock price once the lockup period expires. In terms of economic magnitude, for example, the coefficients in column (5) indicate that a one standard deviation increase in the dispersion measure implies a 44.96% decrease from the mean value of the three-month BHRs. Finally, the results in columns (7) to (9) suggest that bidders' divergent opinions can predict BHRs one

¹⁸ We also control for firm- and issue-level characteristics but omit the coefficients here to save space.

month post issue but that the effect, with a t-value of -1.52, is not significant. This finding confirms that divergent opinions of bidders without the three-month lockup provision can be immediately reflected in stock returns. In unreported analysis, we obtain similar results using CARs as measures of stock performance. Overall, then, consistent with the *uncertainty resolution* channel, our results strongly suggest that premarket divergence of bidder opinions does contain private information about post-issue stock performance, whose rapid impoundment into stock prices is hindered by the lock-up provision.

B. Dispersion and Post-IPO Operating Performance

We next examine whether our dispersion measure can also forecast long-run post-IPO firm operating performance and, if so, how. Following previous studies, we proxy such performance by one-, two-, and three-year post-issue returns on equity (ROE) and returns on assets (ROA). According to the descriptive statistics in Panel A of Table X, firm operating performance declines over the three-year post-issue period, with ROE dropping from 8.21% in the one-year post-issue period to 7.68% two years post issue, a 6.46% reduction. Consistent with Jain and Kini (1994) and Purnanandam and Swaminathan (2004), we also find that issuing firms exhibit a decline in post-issue operating performance relative to their pre-issue levels. To better assess this decline and identify the relation between it and premarket bidder opinion divergence, we use the baseline specification to perform regressions with corresponding one-, two-, and three-year ROE and ROA as dependent variables while controlling for the same holding period stock market index returns. The variable of interest is the dispersion measure *STD*, and we also include industry and year fixed effects for all regressions. As shown in Panel B of Table X, consistent with prior expectation, the coefficient estimates in specifications (1) and (4) indicate a strongly significant and negative relation between the one-year post-issue operating performance as measured by ROE and ROA and the premarket divergence of opinions among bidders.

****** Insert Table X about here ******

In addition to being statistically significant, our findings are meaningful in terms of economic magnitude. For example, the estimates for specification (1) reveal that a one standard deviation increase in our dispersion measure translates into a 2.54% decrease in firm operating performance relative to the average one-year post issue operating performance measured by ROE. We also find not only that the predictive power of divergent bidder opinions is weakened and only marginally significant for the two-year post-issue operating performance measures but the

relation totally disappears for three-year post-issue performance (specifications (3) and (6)). Taken together, our results suggest that bidder opinion divergence (as measured by bidding price dispersion) has strong predictive power mostly for one-year post-issue firm operating performance.

C. *Bidder Characteristics and Bid Price Accuracy*

To examine whether and how institutional bidder characteristics affect bid price accuracy, thereby identifying the informed bidders in an IPO, we measure bid price inaccuracy as the percentage difference between the bid price and final offer price. The first bidder characteristic that may influence IPO pricing is the geographic distribution of both bidders and IPO firms, as suggested by the tendency for local investors to be better informed about a firm's prospectus than nonlocal investors. For example, Baik, Kang, and Kim (2010), find that local investors outperform nonlocal investors in informed trading by exploiting their informational advantages. On the other hand, Hong, Kubik, and Jeremy (2005) demonstrate that local investors behave similarly toward a particular stock within the same time period, even when the company of interest is located far away. This latter finding suggests that information about firm quality may also be spread by word of mouth over a geographically interconnected investor network. Recognizing both possibilities, we use the mean distances between the bidder and IPO firm and between the bidder and other bidders to proxy for the extent of private information a bidder has about the IPO. We expect the distance measure to be positively correlated with bid price inaccuracy. Following Cornelli and Goldreich (2001), we also test other bid characteristics, including bid size, timing, and bidder type and frequency of participation, each of which has separate implications for IPO pricing efficiency.

According to the extant literature, large bidders are better informed and large bids are favored by underwriters in IPO allocation (Cornelli and Goldreich, 2001; Aggarwal, Prabhala and Puri, 2002). Late bids might be more informative than early bids because of the time needed for information spillovers to materialize. We therefore include a dummy variable indicating whether the bid is submitted relatively early or not. We also create a dummy variable to proxy for regular bidders who may be better at pricing IPOs either because of greater pricing experience or more precise private information gleaned from their close business relations with the underwriters. Additionally, because bidders with strong in-house equity research departments (brokerage and fund management firms) and superior information acquisition and production

skills may predict the IPO offer price more precisely and bid more wisely, we introduce a bidder type dummy that explicitly controls for the effect of investor type heterogeneity on IPO pricing. We then identify determinants of bid price inaccuracy by running different specifications of the following regression:

$$\begin{aligned} \text{Bid price inaccuracy} = & \alpha + \beta_1 * \text{Distance} + \beta_2 \text{Largest bid} + \beta_3 \text{Second largest bid} \\ & + \beta_4 \text{Early bid} + \beta_5 \text{Late bid} + \beta_6 \text{High frequency} \quad (2) \\ & + \beta_7 \text{Medium frequency} + \beta_8 \text{Bidder type} + \varepsilon \end{aligned}$$

Here, the dependent variable is *Bid price inaccuracy*, defined as the percentage deviation between individual bid prices and the final offer price. For invalid bids with a bid price below the offer price, we measure bid price inaccuracy as the maximum price deviation among all bids within an IPO. Several independent variables capture different bidder characteristics: *Distance* is constructed as the natural logarithm of the arithmetic mean of the flight distance between the IPO firm and the bidder and the average flight distance between the bidder and all other bidders as a group. We also directly calculate the Euclidian distance between bidder and IPO firm and among bidders based on their geographic coordinates as well as flight distance. *Largest bid* (*Second largest bid*) is a dummy equal to one if the bid size is in the fourth (third) size quartile for that IPO, and *early bid* (*late bid*) is a dummy equal to one if the bid submission time falls in the first (fourth) quartile. We also split bidders into three categories based on the frequency distribution of their past IPO participation, with *High frequency* (*Medium frequency*) equal to one if the total number of IPOs participated in is in the third (second) tercile during our full sample period. Finally, *Bidder type* again categorizes bidders into domestic brokerage firms (BF), fund management firms (FM), financial firms (FF), trust companies (TC), insurance companies (IC), and qualified foreign institutional investors (QF). For this bid level analysis, however, we add two more bidder types: qualified large individual bidders (ID) and institutional bidders (II) independently recommended by the lead underwriter. It should be noted that our earlier analysis (Section IV.B) does not assess the impact of these two groups' divergent opinions on IPO underpricing because dispersion measures for these categories are very rare at the IPO level. Finally, to alleviate concerns that the above variables may not capture unique issue characteristics, the regression also includes issue level fixed effects with standard errors clustered at this level.

Panel A of Table XI reports the estimates of the OLS regressions in which the dependent variable is the continuous measure of bid price inaccuracy defined earlier. Consistent with our expectations, both the primary and alternative distance measures are positively and significantly correlated with bid price inaccuracy at the 1% level—meaning more distant bidders have less accurate bids. This finding suggests that geographic proximity between bidder and IPO firm, as well as between bidder and other bidders as a group, facilitates bidder information acquisition and/or production, thereby helping them improve IPO pricing efficiency. In terms of economic magnitude, all else being equal, the coefficient estimates in column (2) suggest that a one standard deviation decrease in the distance measure leads to a 1.18% ($=0.027 \times 0.436$) improvement in the accuracy of IPO pricing. The coefficients on the two bid size variables are statistically insignificant, confirming that bid size is not a proxy for information about IPO pricing. The coefficient of the dummy variable for early (late) bids, however, is positive (negative) and significant, suggesting that late bids contain more private information about IPO offer price than early bids. As might be expected, both high frequency and medium frequency bidders are better informed than infrequent bidders. We also note interesting differences among bidders from different industries: consistent with the information production or business relation assumption discussed earlier, among those recommended by the lead underwriter, financial firms (FF), fund management firms (FM), trust companies (TC), and institutional investors (II) seem to have a considerable information advantage over insurance companies (IC) and individual investors (ID).

****** Insert Table XI about here ******

We test the robustness of these results using a Probit model in which the dependent variable is a dummy variable indicating whether the bid price is above the offer price or not (see Panel B). Interestingly, the bid characteristics that were statistically significant in the OLS regression continue to be significant in the Probit model. Their signs, however, differ. For example, the proxies for frequent bidders have positive coefficients in Panel A but become negative in Panel B, which seems rational given that an unreasonably high bid price impairs the efficiency of IPO pricing but increases the likelihood of a bid price above the final offer price. Overall, the signs and significances of the estimated coefficients in both the OLS regressions and Probit models strongly suggest that several bidder characteristics, including bidder type and bidder participation frequency, contain private information about IPO pricing.

D. Dispersion and First-day Trading Patterns

We first examine the forecasting power of the divergence of opinions among investors on a variety of IPO related characteristics such as first-day turnover, post-issue return volatility, oversubscription rate in the auction tranche and the likelihood of the closing price falling below the offer price. Specifically, turnover is defined as the proportion of first-day trading volume to the number of IPO shares offered. Return volatility is calculated as the annualized 30-day volatility of market adjusted returns from the IPO listing date. Auction tranche oversubscription is defined as the natural logarithm of the ratio of total bid quantities in the auction tranche from institutional investors divided by the total number of shares offering in the auction tranche. The Offer-Break dummy is an indicator variable that takes a value of one if the first-day closing price falls below the offer price, and otherwise 0. As discussed before, the premarket divergence of opinions among bidders captures a unique component of *ex ante* uncertainty about an issue. Following this logic, we expect that issues with a higher level of divergence of opinion exhibit both greater first-day turnover and greater one-month return volatility. In contrast, we expect that the dispersion measure should negatively impact the auction tranche oversubscription rate in the sense that valuation uncertainty and estimation risk will reduce market demand. Also, we expect the dispersion measure to have a negative impact on the likelihood of offer price breaks since the binding short-sale constraints on the first trading day will have a tendency to raise the closing price *ex-ante*.

Table XII reports regression results on the predictive power of the premarket divergence of opinions among bidders for subsequent IPO first-day turnover, one-month return volatility, auction tranche oversubscription rate and the likelihood of closing price below offer price, respectively. As before, we use the same set of control variables and include industry and year fixed effects for all regressions. Specifications (1) to (3) use ordinary least squares (OLS) regressions while Specification (4) employs a Probit model using the Offer-Break dummy as dependent variable. Consistent with expectations, results reveal that divergence of opinions among investors is positively and significantly related to first-day turnover and post-issue return volatility. Our results also indicate that, on average, issues with greater divergence of opinions about their quality are less oversubscribed from auction tranche bidders and are less likely to have an offer price break. In terms of economic magnitude, we find that a one standard deviation increase in the dispersion measure increases the first-day turnover and one month post-issue

return volatility by 2.39% and 8.85% from their mean values, respectively. Likewise, holding other control variables constant, a one percent increase in the dispersion measure decreases the oversubscription rate by 2.71% and reduces the likelihood of offer price breaks by 0.79%. This is both economically and statistically significant.

****** Insert Table XII about here ******

In sum, these findings suggest that our measure of premarket divergence of opinions among bidders also has strong predictive power in determining other IPO related characteristics, such as first-day turnover ratio, one-month post-issue return volatility, auction tranche oversubscription rate and the likelihood of offer price breaks.

V. Conclusions

This study explores the information content of institutional bids using a large and proprietary sample of Chinese IPO data that contains unique and detailed information on bids and allocations from July 2009 through November 2012. We find that institutional bidding dispersion is positively related to IPO first-day returns by discounting the offer price as a compensation for investors' bearing valuation uncertainty and estimation risk. Further, institutional pre-market bidding dispersion is negatively predictive of IPO firms' operating performance one year post IPO and stock returns three months post IPO, suggesting that the bids of institutional investors carry private information about IPO valuations. Our baseline results remain robust for alternative dispersion measures. The findings are not driven by endogeneity problems since the IV regressions that control for the endogeneity problem by instrumenting bidder opinion divergence with the ratio of the lead underwriter's IPO valuation range to its midpoint as the instrumental variable yield similar results. Further the findings remain robust in the tests with two natural experiments of November 2010 and May 2012 regulatory changes in the IPO auction process. These two IPO reforms are shown to have a significant impact on bidding dispersion: in the post-reform period, institutional investors bid with more information, resulting in a stronger relation between institutional dispersion and IPO underpricing. Our research thus not only sheds new light on the efficiency of the IPO allocation process in China but highlights the importance of institutional factors and the effects of regulatory reforms in the IPO market in China, the world's largest emerging economy.

References

- Abel, A., 1989. Asset prices under heterogeneous beliefs: implications for the equity premium. Unpublished working paper, University of Pennsylvania.
- Aggarwal, R., Prabhala, N. R., Puri, M., 2002. Institutional allocation in initial public offerings: Empirical evidence. *Journal of Finance* 57, 1421-1442.
- Baik, B., Kang, J., Kim, J., 2010. Local institutional investors, information asymmetries, and equity returns. *Journal of Financial Economics* 97, 81-106.
- Banerjee, S. Dai, L., Shrestha, K., 2011. Cross-country IPOs: What explains differences in underpricing? *Journal of Corporate Finance* 17, 1289-1305.
- Barniv, R., Myring, M. J., Thomas, W. B., 2005. The association between the legal and financial reporting environments and forecast performance of individual analysts. *Contemporary Accounting Research* 22, 727-758.
- Barron, O., Kim, O., Lim, S., Stevens, D., 1998. Using analysts' forecasts to measure properties of analysts' information environment. *The Accounting Review* 73, 421-433.
- Barron, O., Stuerke, P., 1998. Dispersion in analyst's earnings forecasts as a measure of uncertainty. *Journal of Accounting, Auditing, and Finance* 13, 243-268.
- Barry, C. B. Brown, S. J., 1986. Limited information as a source of risk, *Journal of Portfolio Management*, 12, 66-72,
- Beatty, R. Ritter, J., 1986. Investment banking, reputation, and the underpricing of initial public offerings. *Journal of Financial Economics* 15, 213-232.
- Benveniste, L. M. Spindt, P. A., 1989. How investment bankers determine the offer price and allocation of new issues. *Journal of Financial Economics* 24, 343-361.
- Benveniste, L.M. Wilhelm, W.J., 1990. A comparative analysis of IPO proceeds under alternative regulatory environments. *Journal of Financial Economics* 28, 173-207.
- Berkman, H., Cole, R. A., Fu, L. J., 2010. Political connections and minority-shareholder protection: Evidence from securities-market regulation in China. *Journal of Financial and Quantitative Analysis* 45, 1391-1417.
- Bushee B J. The influence of institutional investors on myopic R&D investment behavior. *Accounting Review*, 73, 305-333
- Cai J., He J., Zhai W., 2015. Do fools rush in? IPOs and investor sophistication. 64th Annual Meeting of Midwest Finance Association.
- Carter, R., Manaster, S., 1990. Initial public offerings and underwriter reputation. *Journal of Finance* 45, 1045-1067.
- Chan, K., Wang, J., Wei, K. C., 2004. Underpricing and long-term performance of IPOs in China. *Journal of Corporate Finance* 10, 409-430.
- Chen, C., Shi, H., Xu, H., 2014. The IPO underwriting market share in China: Do ownership and quality matter? *Journal of Banking and Finance* 46, 177-189.
- Chiang, Y. M., Qian, Y., Sherman, A. E., 2010. Endogenous entry and partial adjustment in IPO auctions: Are institutional investors better informed? *Review of Financial Studies* 28, 1200-1230.
- Chen, G., Firth, M., Kim, J. B., 2004. IPO underpricing in China's new stock markets. *Journal of Multinational Financial Management*, 14, 283-302.
- Chen, G., Firth, M., Xu, L. P., 2009. Does the type of ownership control matter? Evidence from China's listed companies. *Journal of Banking and Finance*, 33, 171-181.
- Chen, C., Shi, H., Xu, H., 2014. The IPO underwriting market share in China: Do ownership and quality matter? *Journal of Banking and Finance* 46, 177-189.
- Chemmanur, T. J., 1993. The pricing of initial public offerings: A dynamic model with information production. *Journal of Finance* 48, 285-304.
- Chemmanur, T. J., He, S., Hu, G., 2009. The role of institutional investors in seasoned equity offerings. *Journal of Financial Economics* 94, 384-411.
- Chemmanur, T. J., Hu, G., Huang, J., 2010. The role of institutional investors in initial public offerings. *Review of Financial Studies* 23, 4496-4540.

- Chi, J., Padgett, C., 2005. The performance and long-run characteristics of the Chinese IPO market, *Pacific Economic Review* 10, 451-469.
- Choi, S.D., Lee, I., Megginson, W. L., 2010. Do privatization IPO firms outperform in the long run? *Financial Management* 39, 153-185.
- Cornelli, F., Goldreich, D., 2001. Book building and strategic allocation. *Journal of Finance* 56, 2337-2369.
- Cornelli, F., Goldreich, D., 2003. Bookbuilding: How informative is the order book? *Journal of Finance* 58, 1415-1443.
- Cornett, M. M., Marcus, A. J., Saunders, A., Tehranian, H., 2007. The impact of institutional ownership on corporate operating performance. *Journal of Banking & Finance* 31, 1771-1794.
- Degeorge, F., Derrien, F., Womack, K.L., 2010. Auctioned IPOs: the US evidence. *Journal of Financial Economics* 98, 177-194.
- Derrien, F., 2005. IPO pricing in “Hot” market conditions: Who leaves money on the table? *Journal of Finance* 60, 487-521.
- Diether, K., Malloy, C., Scherbina, A., 2002. Difference of opinion and the cross-Section of Stock Returns. *Journal of Finance* 57, 2113-2141.
- Dorn, D., 2009. Does sentiment drive the retail demand for IPOs? *Journal of Financial and Quantitative Analysis* 44, 85-108.
- Epstein, L. G., Wang, T., 1994. Intertemporal asset pricing under Knightian uncertainty. *Econometrica* 62, 283-322.
- Fernando, C., Gatchev, V., May, A., Megginson, W. 2015. Prestige without purpose? Reputation, differentiation, and pricing in U.S. equity underwriting, *Journal of Corporate Finance* 32, 41-63.
- Field L. C., Lowry, M., 2009. Institutional versus individual investment in IPOs: The importance of firm Fundamentals. *Journal of Financial and Quantitative Analysis* 44, 489-516.
- Firth, M., Lin, C., Zou, H., 2010. Friend or foe? The role of the state and mutual fund ownership in the split shares structure reform in China. *Journal of Financial and Quantitative Analysis* 45, 685-706.
- Garfinkel, J. A., Sokobin J., 2006. Volume, opinion divergence, and returns: A study of post-earnings announcement drift. *Journal of Accounting Research* 44, 85-112
- Gaspar J. M., Massa M., Matos P., 2006. Favoritism in mutual fund families? Evidence on strategic cross-fund subsidization. *Journal of Finance* 61, 73-104.
- Gillan, S. L., Starks, L. T., 2000. Corporate governance proposals and shareholder activism: The role of institutional investors. *Journal of Financial Economics* 57, 275-305.
- Goldstein M A, Irvine P, Puckett A., 2011. Purchasing IPOs with commissions. *Journal of Financial and Quantitative Analysis* 46, 1193-1225.
- Hanley, K., Wilhelm, W.H., Jr., 1995. Evidence on the strategic allocation of initial public offerings. *Journal of Financial Economics*, 37, 239-257.
- Hong, H., Kubik, J., Jeremy, J. C., 2005. Thy neighbor’s portfolio: Word-of-mouth effects in the holdings and trades of money managers. *Journal of Finance* 60, 2801-2824
- Houge, T., Loughran, T., Suchanek, G., Yan, X., 2001. Divergence of opinion, uncertainty and the quality of initial public offerings. *Financial Management* 30, 5-23.
- Imhoff, E., Lobo, G., 1992. The effect of ex ante earnings uncertainty on earnings response coefficients. *Accounting Review* 67, 427-439.
- Jacob, J., Lys, T. Z., Neale, M. A., 1999. Expertise in forecasting performance of security analysts. *Journal of Accounting and Economics* 28, 51-82.
- Jagannathan, R., Jirnyi, A. Sherman, A.G., 2015. Share auctions of initial public offerings: Global evidence. *Journal of Financial Intermediation* 24, 283-311.
- Jain, B. A., Kini, O., 1994. The post-issue operating performance of IPO firms. *Journal of Finance* 49, 1699-1726.
- Jenkinson, T., Jones, H., 2004. Bids and allocations in European IPO book building. *Journal of Finance* 59, 2309-2338.
- Johnson, T. C., 2004. Forecast dispersion and the cross-section of expected returns. *Journal of Finance* 59, 1957-1978.

- Kandel, S., Sarig, O., Wohl, A., 1999. The demand for stocks: An analysis of IPO auctions. *Review of Financial Studies* 12, 227-247.
- Ke, B., Ramalingegowda, S., 2005. Do institutional investors exploit the post-earnings announcement drift? *Journal of Accounting and Economics* 39, 25-53.
- Li, B., Megginson, W., Shen, Z., Sun, Q., 2016. Do share issue privatizations really improve firm performance in China? Working paper (University of Oklahoma).
- Li, K., Wang, T., Cheung, Y. L., Jiang, P., 2011. Privatization and risk sharing: Evidence from the split share structure reform in China. *Review of Financial Studies* 24, 2499-2525.
- Liu, S., 2014. Performance persistence of institutional investors in IPO market: Evidence from China. Master's thesis (Lingnan University, Hong Kong).
- Ljungqvist, A., 2004. IPO Underpricing, *Handbook in Corporate Finance: Empirical Corporate Finance*, edited by B. Espen Eckbo.
- Ljungqvist, A. P., Jenkinson, T., Wilhelm, W. H., Jr., 2003. Global integration in primary equity markets: The role of U.S. banks and U.S. investors. *Review of Financial Studies* 16, 63-99.
- Ljungqvist, A., Wilhelm, W. J., 2003. IPO pricing in the Dot-Com Bubble. *Journal of Finance* 58, 723-752.
- Lowry, M., Officer, M.S., Schwert, G.W., 2010. The variability of IPO returns. *Journal of Finance* 65, 425-465.
- McGuinness, P., 1992. An examination of the underpricing of initial public offerings in Hong Kong: 1980-1990. *Journal of Business Finance and Accounting* 19, 165-186.
- Megginson, W., 2015. Privatization major deals and trends in 2013 and 2014. *Privatization Barometer 2014 Report* (FEEM, Milan).
- Megginson W. L., Weiss, K. A., 1991. Venture capitalist certification in initial public offerings. *Journal of Finance* 46, 879-903.
- Merton, R. C., 1987. A simple model of capital market equilibrium with incomplete information. *Journal of Finance* 42, 483-510.
- Miller E. M., 1977. Risk, Uncertainty, and Divergence of Opinion. *Journal of Finance* 32, 1151-1168.
- Mok, M. K., Hui, Y. V., 1998. Underpricing and after-market performance of IPOs in Shanghai, China. *Pacific-Basin Finance Journal* 6, 453-474.
- Neupane, S., Poshakwale, S.S., 2012. Transparency in IPO mechanism: Retail investors' participation, IPO participation, IPO pricing and returns. *Journal of Banking and Finance* 36, 2064-2076.
- Nimalendran M., Ritter J. R., Zhang D., 2007. Do today's trades affect tomorrow's IPO allocations? *Journal of Financial Economics* 84, 87-109.
- Petersen, M., 2009. Estimating Standard Errors in Finance Panel Data Sets: Comparing Approaches. *Review of Financial Studies* 22, 435-480.
- Purnanandam, A.K., Swaminthan, B., 2004. Are IPOs really underpriced? *Review of Financial Studies* 17, 811-847.
- Reuter J., 2006. Are IPO allocations for sale? Evidence from mutual funds. *Journal of Finance* 61, 2289-2324.
- Ritter, J., 1984. The 'hot issue' market of 1980. *Journal of Business* 32, 215-240.
- Ritter, J., 2011. Equilibrium in the Initial Public Offerings Market. *Annual Review of Financial Economics* 3, 347-374.
- Ritter, J., Welch, I., 2002. A review of IPO activity, pricing and allocation. *Journal of Finance* 57, 1795-1828.
- Ritter J. R., Zhang D., 2007. Affiliated mutual funds and the allocation of initial public offerings. *Journal of Financial Economics* 86, 337-368.
- Roberts, M.R., Whited, T., 2012. Endogeneity in empirical corporate finance. In: Constantinides, G., Harris, M., Stulz, R. (eds.), *Handbook of the Economics of Finance*, Elsevier, vol. 2.
- Schnitzlein, C R. Shao, M., and Sherman, A E., 2015. Come on in, the water's fine! An experimental examination of hybrid IPO auctions with a public pool. Working paper, University of Vermont.
- Sherman, A.E., 1992. The pricing of best efforts new issues. *Journal of Finance* 47, 781-790.
- Sherman, A.E., 2005. Global trends in IPO methods: Book-building versus auctions with endogenous entry. *Journal of Financial Economics* 78, 615-644.
- Varian, H. R., 1985. Divergence of opinion in complete markets: A note. *Journal of Finance* 40, 309-317.

Yu, T., Tse, Y.K., 2006. An empirical examination of IPO underpricing in the Chinese A-share market. *China Economic Review* 17, 363-382.

Appendix A: Variable Definitions

Variables	Definition
Panel A: Measure of dispersion	
Dispersion Measure(STD)	Defined as the ratio of the cross-sectional standard deviation of institutional investors' pre-IPO bidding price to the mean bidding price.
Dispersion Measure(STD weighted)	Dispersion in institutional investors' bidding price weighted by their bidding volumes.
Dispersion Measure(MAD)	Defined as mean absolute deviation of bidding price surrounding the mean bidding price scaled by the mean bidding price.
Panel B: IPO and firm characteristics	
First-day Return	Defined as the ratio of the closing price of the stock on its first trading day less the offer price to its offer price.
Turnover	The proportion of trading volume to the number of IPO shares.
Offer-Break dummy	An indicator variable equal to one if the IPO first day closing price falls below (i.e., offer price breaks) the offer price.
Return Volatility	Annualized 30-day volatility of market adjusted returns from the IPO listing date.
Firm age	Log (# of years between founding and offering).
Firm size	Log (Pre-issue book value of total assets in Millions RMB).
Offer size	Log (# of shares offered times offer price in Millions RMB).
Time gap	Log (1+ # of days between listing and offering).
ROE	Return on equity calculated for the latest fiscal year prior to IPO.
market return	One month market return prior to IPO.
Oversubscription	Log (ratio of total bid quantities in auction tranche from institutional investors divided by the total number of shares offering in auction tranche).
Log # Institutions	Log (1+ # of institutional participants).
VC/PE dummy	An indicator variable equal to one if the firm is backed by VC or PE, and zero otherwise.
ReguChange1	An indicator variable equal to one if the IPO bidding date is later than November 5, 2010 when the CSRC conducted the second round IPO reform.
ReguChange2	Indicator variable equal to one if IPO listing date is after May 25, 2012 when CSRC conducted the third round IPO reform.
ChiNext dummy	Indicator variable equal to one if the firm is listed in the Shenzhen ChiNext Board and zero if listed in the SME Board.
Lead reputation (#)	Log (1+total number of IPOs the lead underwriter has managed prior to the current IPO).
Panel C: Bidder and bid characteristics	
Bid price inaccuracy	Percentage difference between bid price and offer price.
Distance	Log (average flight distance between bidder and IPO firm and average flight distance between bidder and all other bidders).
Distance (alternative)	Log (average of Euclidian distance between bidder and IPO firm and average Euclidian distance between bidder and all other bidders based on their geographical coordinates)
Largest (2 nd largest) bid	A dummy variable that takes on a value of one if the bid size is in the fourth (third) size quartile within an IPO.
Early (Late) bid	A dummy variable that equals to one if the bid submission time falls in the first (fourth) quartile within an IPO.
High (Medium) frequency	A dummy variable set to one if total # IPOs the bidder participated in is in the third (second) quartile during sample period.
Bidder type	Dummy variables indicating bidder type: Brokerage firms (BF), fund management (FM), financial firms (FF), trust companies (TC), insurance companies (IC), qualified foreign institutional investors (QF), individual bidders (ID) and institutional bidders (II)independently recommended by the lead underwriter.

Table I: Descriptive Statistics. This table summarizes our IPO sample, which consists of 783 Chinese IPOs listed on the Shenzhen SME Board or the Shenzhen ChiNext Board between 10 July 2009 and 2 November 2012. Panel A reports mean, median, standard deviation, 10th, 25th, 75th, and 90th percentile of the main variables used in this paper. Panel B reports the Spearman correlation matrix, where ***, **, and * indicate 1%, 5%, and 10% statistical significance levels, respectively. Panel C reports the reports mean, median, standard deviation, 10th, 25th, 75th, and 90th percentile of the free-riding level.

Panel A: Summary Statistics

Variable	N	Mean	Median	SD	P10	P25	P75	P90
First-day return	783	0.3684	0.2719	0.4540	-0.0374	0.0768	0.5236	0.8547
Turnover	783	0.7101	0.7700	0.2039	0.3600	0.6300	0.8500	0.8900
STD	783	0.1557	0.1516	0.0336	0.1172	0.1330	0.1737	0.2022
MAD	783	0.1228	0.1202	0.0278	0.0903	0.1041	0.1383	0.1607
Firm age (Log)	783	1.8899	2.1058	0.7520	0.7747	1.3641	2.4122	2.7188
Offer size (Log)	783	6.4992	6.4378	0.6049	5.8171	6.0808	6.8416	7.2714
Time gap (Log)	783	2.4850	2.4849	0.2515	2.1972	2.3026	2.6391	2.7726
ROE	783	0.2370	0.2237	0.1220	0.1113	0.1620	0.2887	0.3632
Index return %	783	-0.2970	-0.3088	7.5257	-8.8330	-6.0113	4.8580	9.6056
Oversubscription (Log)	783	3.3804	3.4898	1.1758	1.7750	2.5055	4.3037	4.8250
Log # institutions	783	4.1662	4.1744	0.4636	3.5264	3.8067	4.5433	4.7791
VC/PE dummy	783	0.4994	0	0.5003	0	0	1	1
ReguChange1	783	0.5236	1	0.4998	0	0	1	1
ReguChange2	783	0.0830	0	0.2761	0	0	0	0
ChiNext dummy	783	0.4534	0	0.4981	0	0	1	1
Lead reputation	783	3.5230	3.9703	1.2949	1.6094	2.5649	4.5850	4.8283

Panel B: Spearman Correlation Matrix

ID	Variable	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	First-day return	1.00													
2	STD	-0.01	1.00												
3	Firm age	-0.05	-0.08**	1.00											
4	Offer size	-0.27***	-0.05	-0.14***	1.00										
5	Time gap	0.16***	0.15***	-0.18***	0.09**	1.00									
6	ROE	-0.20***	0.01	-0.05	0.05	-0.16***	1.00								
7	Index return %	0.47***	-0.06	-0.01	0.03	0.12***	-0.14***	1.00							
8	Oversubscription	0.48***	-0.13***	-0.25***	0.06*	0.30***	-0.25***	0.13***	1.00						
9	Log # institutions	0.44***	-0.17***	-0.24***	0.23***	0.25***	-0.27***	0.16***	0.88***	1.00					
10	VC/PE dummy	-0.02	-0.09***	-0.02	0.02	-0.02	-0.07**	0.04	-0.04	-0.03	1.00				
11	ReguChange1	-0.37***	-0.15***	0.35***	-0.19***	-0.47***	0.21***	-0.22***	-0.73***	-0.68***	0.06*	1.00			
12	ReguChange2	-0.16***	-0.07**	0.17***	-0.19***	-0.12***	0.12***	-0.09***	-0.13***	-0.14***	0.04	0.29***	1.00		
13	ChiNext dummy	-0.03	-0.10***	0.00	-0.27***	-0.03	0.16***	0.00	-0.09**	-0.19***	0.14***	0.14***	0.06*	1.00	
14	Lead reputation	-0.13***	-0.12***	0.08**	0.05	-0.11***	0.06*	-0.03	-0.22***	-0.18***	0.13***	0.24***	0.10***	0.01	1.00

Panel C: Free-riding Behavior Dynamics

	Free-riding level	N	Median	SD	P10	P25	P75	P90	Mean	Mean Diff
(1)	Full sample	783	0.0076	0.1643	0.0000	0.0000	0.0839	0.2931	0.0854	
(2)	ReguChange1=0	373	0.0645	0.1931	0.0033	0.0157	0.2089	0.4891	0.1514	
(3)	ReguChange1=1	410	0.0000	0.1005	0.0000	0.0000	0.0000	0.0466	0.0254	-0.1259*** (3) vs. (2)
(4)	ReguChange1=1, ReguChange2=0	345	0.0000	0.1090	0.0000	0.0000	0.0000	0.0580	0.0297	
(5)	ReguChange2=1	65	0.0000	0.0110	0.0000	0.0000	0.0000	0.0000	0.0029	-1.98** (5) vs. (4)

Figure 1: Number of IPOs and Average Underpricing. This figure depicts the number of IPOs and average underpricing on a quarterly basis for our full IPO sample which consists of 783 firms listed on the Shenzhen SME board or the ChiNext board during the period of 2009Q3 to 2012Q4. The blue bars show the number of IPOs for each quarter and the red line shows the quarterly average IPO underpricing.

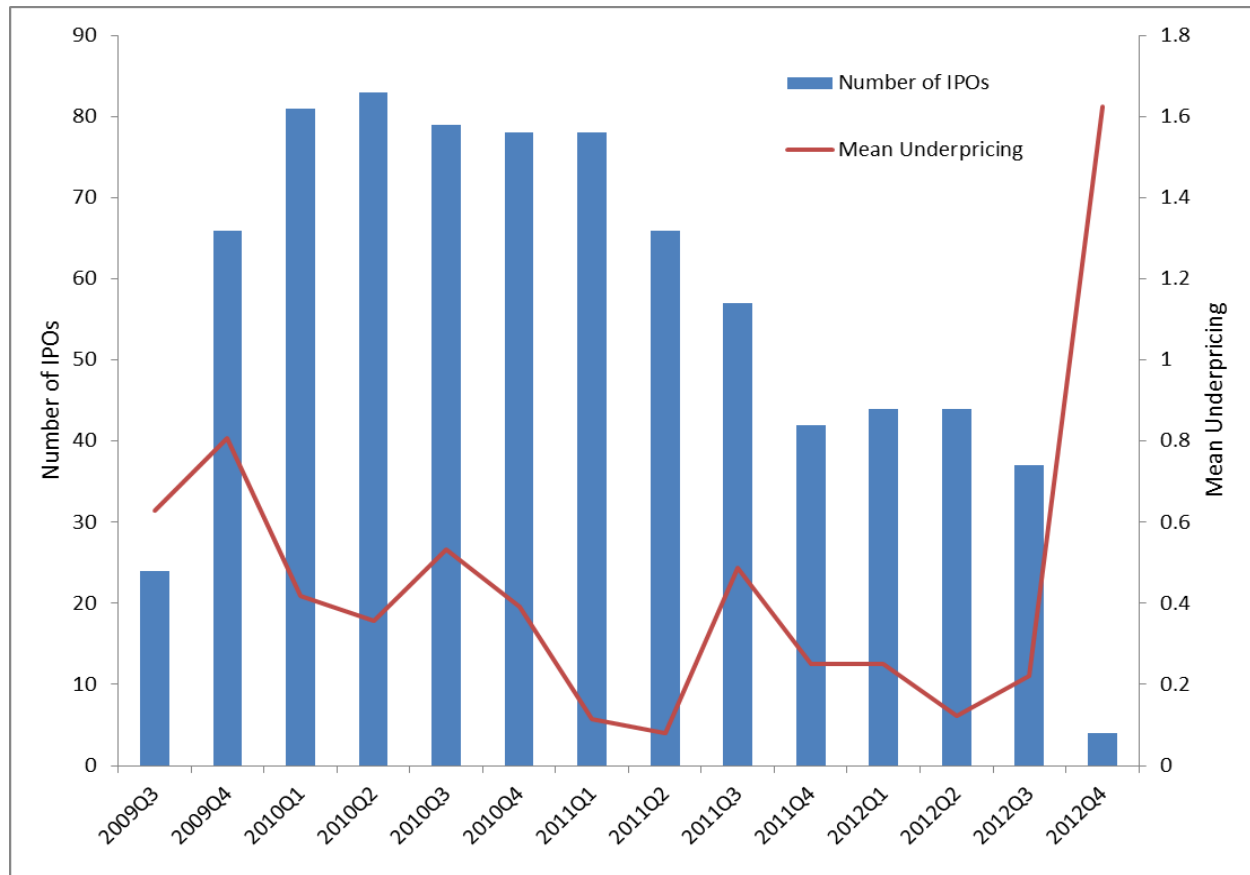


Table II: Impact of Opinion Dispersion on Offer Price, First-day Closing Price and Underpricing – Baseline Regression. The dependent variables are the offer price (1), first-day closing price (2) and IPO first-day return (3), defined as the ratio of the closing price of the stock on its first trading day less the offer price, divided by its offer price. The variable of interest is dispersion measure *STD*, defined as the ratio of the cross-sectional standard deviation of institutional investors' bidding price in the auction bidding process to the mean bidding price. Variable definitions are given in Appendix A. Reported are coefficients and *t*-values calculated using the industry clustered standard errors. ***, **, and * indicate 1%, 5%, and 10% statistical significance levels, respectively.

	Offer price (1)	First-day Closing price (2)	Underpricing (3)
STD	-24.4284* (-2.18)	-10.8202 (-1.09)	1.5737*** (4.93)
Firm age	0.7853 (1.23)	1.2757* (2.08)	0.0079 (0.77)
offer size	12.2120*** (19.88)	11.2396*** (12.84)	-0.2129*** (-15.36)
Time gap	-0.3808 (-0.31)	1.8457 (0.68)	0.1043*** (3.56)
ROE	28.4994*** (25.00)	39.4188*** (17.13)	0.0245 (0.86)
market return	-0.0322** (-2.52)	0.4541*** (19.93)	0.0213*** (16.64)
Oversubscription	-0.1422 (-0.56)	1.9116*** (5.04)	0.1414*** (6.31)
Log # institutions	4.5183*** (13.32)	9.0264*** (12.18)	0.1310*** (6.31)
VC/PE dummy	0.0655 (0.06)	0.1107 (0.12)	-0.0071 (-0.34)
ReguChange1	1.8077** (2.22)	5.9966*** (3.96)	0.2291*** (17.44)
ReguChange2	-0.8586 (-0.88)	-2.7209 (-1.14)	-0.0839 (-1.32)
ChiNext dummy	7.5146*** (13.92)	8.7562*** (18.03)	-0.0676*** (-3.28)
Lead reputation	-0.0845 (-0.40)	-0.3859 (-1.12)	0.0070 (1.60)
Constant	-74.0747*** (-10.20)	-92.3623*** (-6.71)	0.2402 (1.61)
Industry FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
<i>N</i>	783	783	783
adj. <i>R</i> ²	0.446	0.416	0.384

Table III: Dispersion and IPO Underpricing – Instrumental Variable Approach. This table reports results from 2SLS instrumental variable regressions, where the relative ratio of the lead underwriter’s IPO valuation range to the midpoint of the valuation range, $2 \times \frac{Pr_{CH} - Pr_{CL}}{Pr_{CH} + Pr_{CL}}$, is used as an instrumental variable for the bidding price dispersion, *STD*, defined as the ratio of the cross-sectional standard deviation of institutional investors’ bidding price in the auction bidding process to the mean bidding price. The dependent variable is the IPO first-day return, defined as the ratio of the closing price of the stock on its first trading day less the offer price, divided by its offer price. Variable definitions are given in Appendix A. Reported are coefficients and *t*-values calculated using the industry clustered standard errors. ***, **, and * indicate 1%, 5%, and 10% statistical significance levels, respectively.

<i>Panel A: First-stage regression</i>		<i>Panel B: Second-stage regression</i>	
Dependent variable: <i>STD</i>		Dependent variable: First-day return	
Price band ratio	0.0545*** (6.97)	Fitted <i>STD</i>	6.8883*** (5.89)
Firm age	-0.0012** (-2.26)	Firm age	0.0122 (1.10)
Offer size	-0.0042** (-2.46)	Offer size	-0.1882*** (-15.11)
Time gap	0.0123*** (6.11)	Time gap	0.0307 (1.72)
ROE	0.0013 (0.19)	ROE	0.0182 (0.79)
market return	-0.0004*** (-8.09)	market return	0.0235*** (28.70)
Oversubscription	-0.0091*** (-5.58)	Oversubscription	0.1907*** (14.52)
Log # institutions	-0.0246*** (-7.48)	Log # institutions	0.2573*** (6.05)
VC/PE dummy	-0.0020 (-1.37)	VC/PE dummy	0.0044 (0.18)
ReguChange1	-0.0288*** (-10.05)	ReguChange1	0.3829** (10.22)
ReguChange2	0.0001 (0.04)	ReguChange2	-0.0828 (-1.27)
ChiNext dummy	-0.0093*** (-11.36)	ChiNext dummy	-0.0170 (-0.53)
Lead reputation	-0.0025*** (-7.92)	Lead reputation	0.0207*** (3.19)
Constant	0.3057*** (20.35)	Constant	-1.4056*** (-3.52)
Industry FE	Yes	Industry FE	Yes
Year FE	Yes	Year FE	Yes
<i>N</i>	783	<i>N</i>	783
adj. <i>R</i> ²	0.246	adj. <i>R</i> ²	0.376

Table IV: Alternative Dispersion Measures and IPO Underpricing. The dependent variable is the IPO first-day return, defined as the ratio of the closing price of the stock on its first trading day less the offer price, divided by its offer price. The variables of interest are the alternative dispersion measures, *MAD* and *STD weighted*, where *MAD* is defined as the mean absolute deviation of the bidding price surrounding the mean bidding price, scaled by the mean bidding price and *STD weighted* is calculated as the bidding price dispersion among institutional investors weighted by their bidding volume. Variable definitions are given in Appendix A. Reported are coefficients and *t*-values calculated using the industry clustered standard errors. ***, **, and * indicate 1%, 5%, and 10% statistical significance levels, respectively.

	Dependent variable: First-day return			
	(1)	(2)	(3)	(4)
MAD	2.0272*** (4.86)	1.9612*** (4.46)		
STD weighted			1.4750*** (4.84)	1.4530*** (4.71)
Firm age	0.0153 (1.01)	0.0081 (0.79)	0.0160 (1.05)	0.0085 (0.84)
Offer size	-0.2140*** (-14.53)	-0.2128*** (-15.96)	-0.2163*** (-14.33)	-0.2148*** (-15.95)
Time gap	0.1240** (3.06)	0.1044*** (3.51)	0.1280** (3.03)	0.1076*** (3.49)
ROE	-0.0211 (-0.77)	0.0248 (0.90)	-0.0270 (-0.99)	0.0193 (0.63)
market return	0.0211*** (19.86)	0.0213*** (16.65)	0.0210*** (20.06)	0.0213*** (16.89)
Oversubscription	0.1417*** (5.71)	0.1408*** (6.30)	0.1422*** (5.60)	0.1415*** (6.19)
Log # institutions	0.1855*** (4.85)	0.1350*** (6.43)	0.1725*** (4.15)	0.1219*** (5.21)
VC/PE dummy	-0.0036 (-0.15)	-0.0070 (-0.33)	-0.0045 (-0.18)	-0.0083 (-0.38)
ReguChange1	0.1711*** (4.39)	0.2288*** (16.13)	0.1686*** (4.46)	0.2280*** (17.48)
ReguChange2	-0.1158* (-2.13)	-0.0867 (-1.37)	-0.1138* (-2.06)	-0.0831 (-1.30)
ChiNext dummy	-0.0644** (-2.98)	-0.0682*** (-3.30)	-0.0625** (-3.03)	-0.0658*** (-3.29)
Lead reputation	0.0093 (1.65)	0.0077 (1.73)	0.0082 (1.42)	0.0067 (1.48)
Constant	-0.1492 (-1.07)	0.2230 (1.51)	-0.0667 (-0.43)	0.3079** (2.32)
Industry FE	No	Yes	No	Yes
Year FE	No	Yes	No	Yes
<i>N</i>	783	783	783	783
adj. <i>R</i> ²	0.379	0.385	0.377	0.383

Table V: Dispersion and IPO Underpricing – Cross-Sectional Analysis. This table reports regression results using subsamples based on offer size, time gap and lead underwriter reputation respectively. The dependent variable is the IPO first-day return, defined as the ratio of the closing price of the stock on its first trading day less the offer price, divided by its offer price. The variable of interest is dispersion measure *STD*, defined as the ratio of the cross-sectional standard deviation of institutional investors' bidding price in the auction bidding process to the mean bidding price. Variable definitions are given in Appendix A. Reported are coefficients and *t*-values calculated using the industry clustered standard errors. ***, **, and * indicate 1%, 5%, and 10% statistical significance levels, respectively.

	Dependent variable: First-day return					
	Offer Size		Time Gap		Underwriter Reputation	
	Small	Large	Short	Long	Low	High
	(1)	(2)	(3)	(4)	(5)	(6)
STD	1.5751*** (3.91)	0.8507*** (4.97)	2.2730*** (4.11)	0.5274* (2.12)	2.6365*** (4.47)	0.9433*** (3.37)
Firm age	-0.0003 (-0.01)	0.0194*** (3.18)	-0.0126 (-0.73)	0.0139 (1.31)	0.0242 (1.19)	-0.0009 (-0.07)
Offer size	-0.5993*** (-6.92)	-0.0283 (-1.61)	-0.1974*** (-9.83)	-0.2566*** (-6.54)	-0.3038*** (-10.25)	-0.1494*** (-8.48)
Time gap	0.2011*** (7.11)	0.0090 (0.18)	-0.3300*** (-3.61)	0.0774 (0.92)	0.0880 (0.89)	0.0685** (2.44)
ROE	-0.1501** (-2.47)	0.0344 (1.04)	0.0954* (1.90)	-0.0959** (-2.33)	0.2779** (2.46)	-0.1135* (-2.20)
market return	0.0279*** (9.69)	0.0152 (2.59)	0.0219 (24.76)	0.0187*** (9.37)	0.0198*** (11.52)	0.0223*** (16.57)
Oversubscription	0.1635*** (3.28)	0.0952*** (22.91)	0.1683*** (9.72)	0.1082*** (3.57)	0.1924*** (7.75)	0.0625** (2.53)
Log # institutions	0.1632*** (3.61)	0.1273*** (5.06)	0.1575*** (4.40)	0.1459*** (7.18)	0.2057*** (4.82)	0.1621*** (3.22)
VC/PE dummy	-0.0282 (-0.78)	0.0001 (0.01)	-0.0249 (-0.99)	-0.0047 (-0.21)	-0.0414** (-2.62)	0.0140 (0.39)
ReguChange1	0.3685*** (12.07)	0.1123*** (4.11)	0.1739*** (11.14)	0.3263*** (12.48)	0.2638*** (9.02)	0.2298*** (14.65)
ReguChange2	-0.0366 (-0.58)	-0.0823** (-2.89)	-0.0565 (-0.77)	-0.2215*** (-5.47)	-0.0190 (-0.27)	-0.1770** (-3.05)
ChiNext dummy	-0.1425*** (-4.37)	-0.0297 (-1.25)	-0.1257*** (-7.92)	-0.0319 (-1.41)	-0.1521*** (-4.57)	0.0059 (0.28)
Lead reputation	0.0389*** (7.81)	-0.0110** (-2.54)	0.0214*** (4.34)	-0.0077 (-0.47)	0.0589*** (4.51)	-0.0156 (-1.64)
Constant	2.3038*** (4.25)	-0.5130* (-2.11)	0.7093*** (4.63)	0.9742*** (3.94)	-0.2811 (-1.43)	0.3455* (1.89)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
<i>N</i>	393	390	493	290	397	386
adj. <i>R</i> ²	0.428	0.387	0.385	0.407	0.373	0.455

Table VI: Dispersion and IPO Underpricing by Bidder Category. The dependent variable is the IPO first-day return. The variable of interest is the dispersion measure STD, defined as the ratio of the cross-sectional standard deviation of institutional investors' bidding price in the auction bidding process, divided by the mean bidding price. We calculate bidding price dispersion for each different type of institutional investors separately. Specifically, STD_BF is the dispersion measure calculated among domestic brokerage firms; STD_FM is the dispersion measured using only fund management firms; STD_FF is calculated among financial firms; STD_TC refers to the dispersion among trust companies; STD_IC is measured using insurance companies only; and STD_QF is the dispersion measured among qualified foreign institutional investors (QFIIs). Variable definitions are given in Appendix A. Reported are coefficients and t-values calculated using the industry clustered standard errors. ***, **, and * indicate 1%, 5%, and 10% statistical significance levels, respectively.

Dependent variable: First-day return						
	(1)	(2)	(3)	(4)	(5)	(6)
STD_BF	1.4043*** (9.65)					
STD_FM		1.1918*** (4.27)				
STD_FF			0.6083* (2.16)			
STD_TC				0.2672 (1.71)		
STD_IC					0.0745 (0.96)	
STD_QF						0.0486 (0.16)
Firm age	0.0053 (0.45)	0.0089 (0.97)	0.0100 (0.66)	0.0088 (0.86)	-0.0268*** (-3.37)	0.0254 (0.43)
Offer size	-0.2176*** (-16.83)	-0.2111*** (-14.66)	-0.2258*** (-15.95)	-0.2229*** (-17.26)	-0.2289*** (-29.33)	-0.1278 (-0.95)
Time gap	0.1021*** (3.80)	0.1290*** (4.80)	0.1124*** (3.79)	0.1287*** (4.40)	-0.2101*** (-7.59)	0.3824** (2.75)
ROE	0.0314 (1.11)	0.0170 (0.53)	-0.0172 (-0.54)	-0.0104 (-0.37)	0.0813* (1.88)	-0.6813 (-1.34)
market return	0.0214*** (19.44)	0.0212*** (16.65)	0.0210*** (12.76)	0.0208*** (16.72)	0.0233*** (25.83)	0.0300*** (3.29)
Oversubscription	0.1369*** (7.24)	0.1394*** (5.97)	0.1476*** (4.33)	0.1350*** (5.30)	0.1920*** (8.10)	0.3244*** (7.50)
Log # institutions	0.1268*** (6.03)	0.1227*** (6.58)	0.0942** (2.30)	0.1120*** (3.61)	0.0542* (2.06)	0.5607** (2.86)
VC/PE dummy	-0.0062 (-0.29)	-0.0082 (-0.37)	-0.0157 (-0.66)	-0.0227 (-1.03)	-0.0554*** (-4.63)	0.1248 (0.97)
ReguChange1	0.2088*** (18.53)	0.2346*** (15.96)	0.1943*** (9.62)	0.1978*** (14.03)	0.1745*** (10.20)	-0.0018 (-0.01)
ReguChange2	-0.0796 (-1.34)	-0.0917 (-1.54)	-0.1513*** (-4.06)	-0.1261* (-1.84)	0.0282 (0.71)	0.0000 (.)
ChiNext dummy	-0.0848*** (-3.44)	-0.0769*** (-3.73)	-0.0820*** (-3.19)	-0.0710** (-2.84)	0.2287 (1.07)	0.0636 (0.57)
Lead reputation	0.0048 (1.05)	0.0063 (1.71)	0.0044 (0.82)	0.0045 (1.11)	0.0242*** (6.97)	-0.0009 (-0.04)
Constant	0.3552** (2.78)	0.2835 (1.70)	0.5906*** (4.63)	0.5770*** (5.04)	1.1869*** (6.56)	-4.2271* (-2.17)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	783	783	709	730	414	86
adj. R ²	0.384	0.383	0.374	0.377	0.341	0.321

Table VII: Impact of IPO Regulatory Change on Dispersion. This table presents regression results on the impact of two IPO regulatory changes on the divergence of opinions about an IPO among institutional bidders. The dependent variable is the dispersion measure *STD*, which is defined as the ratio of the cross-sectional standard deviation of institutional investors' bidding price in the auction tranche, divided by the mean bidding price. The variables of interest are the two regulatory change dummies: *ReguChange1* and *ReguChange2*. *ReguChange1* equals to one if the IPO bidding date is after than November 5, 2010, when the CSRC conducted the second round IPO reform, and zero before then. *ReguChange2* equals to one if the IPO listing date is after than May 25, 2012, when the CSRC conducted the third round IPO reform, and zero before then. We use baseline regression specifications and variable definitions are given in Appendix A. Reported are coefficients and *t*-values calculated using the industry clustered standard errors. ***, **, and * indicate 1%, 5%, and 10% statistical significance levels, respectively.

	Dependent variable: STD	
	Full Sample	Full sample (matched)
ReguChange1	-0.0289*** (-9.61)	
ReguChange2	-0.0002 (-0.09)	-0.0070** (-2.32)
Firm age	-0.0008 (-1.57)	0.0012 (0.24)
Offer size	-0.0046*** (-3.11)	-0.0026 (-0.85)
Time gap	0.0138** (7.10)	-0.0172 (-0.89)
ROE	0.0012 (0.19)	-0.0736*** (-3.74)
market return	-0.0004*** (-7.09)	0.0006 (1.72)
Oversubscription	-0.0093*** (-5.61)	-0.0078* (-2.13)
Log # Institutions	-0.0238*** (-7.04)	-0.0198* (-2.06)
VC/PE dummy	-0.0022 (-1.52)	-0.0031 (-0.45)
ChiNext dummy	-0.0095*** (-11.03)	0.0019 (0.38)
Lead reputation	-0.0026*** (-8.35)	-0.0063*** (-4.42)
Constant	0.3097*** (21.20)	0.3365*** (3.65)
Industry FE	Yes	Yes
Year FE	Yes	Yes
<i>N</i>	783	130
adj. <i>R</i> ²	0.238	0.091

Table VIII: IPO Regulatory Changes, Dispersion and IPO Underpricing. The dependent variable is the IPO first-day return, defined as the ratio of the closing price of the stock on its first trading day less the offer price, divided by its offer price. The dispersion measure *STD* is defined as the ratio of the cross-sectional standard deviation of institutional investors' bidding price in the auction tranche to the mean bidding price. *ReguChange1* equals to one if the IPO bidding date is after than November 5, 2010, when the CSRC conducted the second round IPO reform, and zero before then. *ReguChange2* equals to one if the IPO listing date is after than May 25, 2012, when the CSRC conducted the third round IPO reform, and zero before then. Column (1) includes interaction term between *Dispersion* measure and the *ReguChange1* in our baseline regression. In column (2), we report baseline regression results using the subsample firms with *before* 2010 reform (*ReguChange1* = 0). Results in column (3) describe regression output for subsample firms *after* 2010 reform (*ReguChange1* = 1). Column (4) includes interaction term between *Dispersion* measure and the *ReguChange2* in our baseline regression. In column (5), we report baseline regression results using the subsample firms with *after* 2010 reform but *before* 2012 reform (*ReguChange1* = 1 and *ReguChange2* = 0). Results in column (6) describe regression output for subsample firms *after* 2012 reform (*ReguChange2* = 1). Variable definitions are given in Appendix A. Reported are coefficients and *t*-values calculated using the industry clustered standard errors. ***, **, and * indicate 1%, 5%, and 10% statistical significance levels, respectively.

	Dependent variable: First-day return					
	Full Sample	ReguChange1 = 0	ReguChange1 = 1	Full sample (matched)	ReguChange1 = 1, ReguChange2 = 0	ReguChange2 = 1
	(1)	(2)	(3)	(4)	(5)	(6)
STD	0.9593** (2.70)	0.9903*** (5.09)	2.2825*** (6.26)	3.6195*** (3.88)	2.1865*** (5.49)	7.0816*** (5.62)
STD × ReguChange1	1.0917*** (4.52)					
STD × ReguChange2				2.2934* (2.08)		
ReguChange1	0.0654 (1.78)					
ReguChange2	-0.0774 (-1.20)	0.0000 (.)	-0.1163 (-1.58)	-0.5464** (-2.64)		
Firm age	0.0070 (0.66)	0.0008 (0.05)	0.0073 (0.79)	0.1004* (1.93)	0.0149 (0.26)	0.2189*** (9.79)
Offer size	-0.2153*** (-16.07)	-0.2332*** (-11.56)	-0.2002*** (-13.15)	0.0070 (0.45)	0.0341*** (7.91)	-0.5496*** (-9.26)
Time gap	0.1125*** (3.81)	0.1481*** (7.13)	-0.0452 (-1.17)	-0.3258* (-2.13)	0.1387 (0.79)	-0.7041*** (-3.61)
ROE	0.0217 (0.78)	-0.1103*** (-4.10)	0.1718*** (5.31)	0.4196 (1.36)	0.8752*** (4.06)	1.1643*** (5.38)
market return	0.0211*** (16.81)	0.0220*** (20.14)	0.0181*** (16.18)	0.0189** (2.68)	0.0113* (2.56)	0.0230** (2.72)
Oversubscription	0.1415*** (6.44)	0.1015*** (6.26)	0.1711*** (7.55)	0.4667*** (5.13)	0.1209 (1.53)	0.6069*** (6.49)
Log #	0.1276*** (6.14)	0.2082*** (6.06)	0.1464*** (4.04)	-0.1597* (-2.22)	0.0143 (0.46)	-0.1293 (-0.64)
Institutions	-0.0049 (-0.23)	-0.0014 (-0.05)	-0.0254 (-1.55)	-0.0287 (-0.98)	0.0884 (1.81)	0.1733*** (3.27)
VC/PE dummy	-0.0049 (-0.23)	-0.0014 (-0.05)	-0.0254 (-1.55)	-0.0287 (-0.98)	0.0884 (1.81)	0.1733*** (3.27)
ChiNext dummy	-0.0700*** (-3.14)	-0.0160 (-0.49)	-0.1364*** (-5.42)	-0.2579*** (-3.82)	-0.0918*** (-5.69)	-0.5855*** (-14.67)
Lead reputation	0.0068 (1.46)	0.0010 (0.16)	0.0229*** (5.81)	0.0691*** (3.64)	-0.0338* (-2.04)	0.1154*** (4.82)
Constant	0.3519** (2.49)	0.2036 (0.93)	0.2748 (1.71)	-0.8529 (-0.94)	-0.7339 (-1.00)	1.5152 (1.45)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
<i>N</i>	783	373	410	130	65	65
adj. <i>R</i> ²	0.385	0.359	0.302	0.336	0.304	0.491

Table IX: Dispersion and Post-IPO Stock Performance. This table reports regression results with post-IPO stock performance measured as the one-, three- and six-month buy and hold returns (BHRs) as dependent variables. BHRs are calculated using monthly stock returns starting from the first month after the IPO listing date. Cumulative abnormal returns (CARs) are measured as the cumulative difference between monthly return of IPO and corresponding monthly return of the value-weighted market index. Panel A presents summary statistics of post-IPO stock performance measures. Panel B reports regression results with corresponding stock performance measures as dependent variables. Specifically, the first three columns report results using all sample IPOs. The last six columns reports results using IPOs with and without three-month lockup provision, respectively. The variable of interest is dispersion measure *STD*, defined as the ratio of the cross-sectional standard deviation of institutional investors' bidding price in the auction bidding process to the mean bidding price. In all regression specifications, we control for firm and issue specific characteristics. However, we don't report their coefficient estimates to save space. Variable definitions are given in Appendix A. Reported are coefficients and *t*-values calculated using the industry clustered standard errors. ***, **, and * indicate 1%, 5%, and 10% statistical significance levels, respectively.

Panel A: Summary Statistics of Post-IPO Stock Performance

Variable	N	Mean	Median	SD	P10	P25	P75	P90
BHR_1m	783	-0.62%	-1.72%	15.81%	-18.42%	-11.31%	8.47%	17.96%
BHR_3m	783	-2.58%	-6.22%	21.84%	-26.51%	-17.63%	9.07%	26.46%
BHR_6m	783	-2.07%	-8.81%	29.72%	-34.37%	-22.77%	14.82%	35.54%
CAR_1m	783	-0.60%	-2.36%	12.84%	-13.67%	-8.05%	4.50%	14.58%
CAR_3m	783	-2.43%	-3.90%	17.77%	-23.13%	-13.85%	7.16%	19.61%
CAR_6m	783	-0.58%	-4.02%	23.85%	-29.20%	-17.56%	13.87%	31.20%

Panel B: Impact of Dispersion on Post-IPO Stock Performance

	Full sample IPOs			IPOs with three-month lockup provision			IPOs without three-month lockup provision		
	BHR_1m	BHR_3m	BHR_6m	BHR_1m	BHR_3m	BHR_6m	BHR_1m	BHR_3m	BHR_6m
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
STD	-0.0108 (-0.11)	-0.2874** (-2.85)	0.0062 (0.04)	0.0224 (0.19)	-0.3457*** (-3.29)	0.0441 (0.42)	-0.2079 (-1.52)	0.4062 (1.07)	-0.0883 (-0.09)
Market return 1m	1.3131*** (30.56)			1.3230*** (33.10)			1.6429*** (5.68)		
Market return 3m		1.2037*** (22.87)			1.1917*** (22.57)			1.3181*** (3.86)	
Market return 6m			1.0864*** (31.45)			1.1041*** (29.82)			1.3820*** (14.85)
Other controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	783	783	783	718	718	718	65	65	65
adj. R ²	0.372	0.407	0.366	0.387	0.414	0.373	0.116	0.271	0.261

Table X: Dispersion and Post-IPO Operating Performance. This table reports regression results with post-IPO operating performance measured using the one-, two- and three-year ROE and ROA as dependent variables. Panel A presents summary statistics of post-IPO operating performance measures. Panel B reports regression results with corresponding operating performance measures as dependent variables. The variable of interest is dispersion measure *STD*, defined as the ratio of the cross-sectional standard deviation of institutional investors' bidding price in the auction bidding process, divided by the mean bidding price. Variable definitions are given in Appendix A. Reported are coefficients and *t*-values calculated using the industry clustered standard errors. ***, **, and * indicate 1%, 5%, and 10% statistical significance levels, respectively.

Panel A: Summary Statistics of Post-IPO Operating Performance

Variable	N	Mean	Median	SD	P10	P25	P75	P90
ROE_1yr	783	8.21%	7.79%	2.89%	5.20%	6.29%	9.73%	11.62%
ROE_2yr	731	7.68%	7.50%	4.70%	3.18%	5.01%	10.16%	12.75%
ROE_3yr	544	6.45%	6.48%	8.55%	1.50%	3.72%	9.72%	13.28%
ROA_1yr	783	6.64%	6.29%	2.48%	3.95%	5.03%	7.88%	9.46%
ROA_2yr	731	6.03%	5.86%	3.69%	2.36%	3.77%	7.99%	10.20%
ROA_3yr	544	4.99%	4.83%	4.61%	1.06%	2.60%	7.32%	10.03%

Panel B: Impact of Dispersion on Post-IPO Operating Performance

	ROE_1yr	ROE_2yr	ROE_3yr	ROA_1yr	ROA_2yr	ROA_3yr
	(1)	(2)	(3)	(4)	(5)	(6)
STD	-0.0621*** (-4.63)	-0.1029 (-1.60)	0.0004 (0.01)	-0.0359*** (-4.29)	-0.0565 (-1.23)	-0.0040 (-0.11)
Firm age	0.0005 (0.90)	0.0036** (2.53)	0.0086*** (6.25)	-0.0003 (-0.65)	0.0027** (2.85)	0.0072*** (10.18)
Offer size	0.0085*** (8.33)	0.0107*** (6.00)	-0.0002 (-0.05)	0.0087*** (11.22)	0.0100*** (9.85)	0.0075*** (4.82)
Time gap	0.0083*** (4.90)	0.0123 (1.60)	0.0423** (6.38)	0.0083** (6.32)	0.0114** (2.22)	0.0299*** (3.76)
ROE	0.0606*** (10.41)	0.0446*** (4.20)	0.0909*** (3.55)	0.0652*** (10.61)	0.0501*** (8.26)	0.0559*** (5.83)
Market return 1yr	0.0087 (1.14)			0.0018 (0.28)		
Market return 2yr		0.0162*** (3.11)			0.0062** (2.70)	
Market return 3yr			0.0228*** (4.08)			0.0100 (1.78)
Oversubscription	0.0005 (0.60)	0.0051*** (3.75)	0.0107*** (4.38)	0.0014* (2.15)	0.0037*** (3.89)	0.0048*** (3.46)
Log # institutions	-0.0125*** (-7.99)	-0.0117 (-1.64)	-0.0091* (-1.81)	-0.0056*** (-4.43)	-0.0018 (-0.42)	0.0006 (0.17)
VC/PE dummy	-0.0047*** (-3.19)	-0.0132*** (-7.27)	-0.0109*** (-5.54)	-0.0040*** (-3.17)	-0.0109*** (-7.15)	-0.0076*** (-5.68)
ReguChange1	-0.0014 (-0.32)	0.0016 (0.84)	0.0308*** (5.22)	-0.0004 (-0.12)	0.0061*** (4.77)	0.0189*** (9.83)
ReguChange2	-0.0053*** (-3.59)	-0.0154** (-2.69)	0.0000 (.)	-0.0046** (-2.71)	-0.0140*** (-4.22)	0.0000 (.)
ChiNext dummy	-0.0068*** (-7.29)	-0.0037 (-1.31)	-0.0124*** (-3.23)	-0.0002 (-0.15)	0.0026 (1.35)	-0.0021 (-0.52)
Lead reputation	0.0010* (1.89)	0.0023 (1.68)	0.0006 (0.33)	0.0011** (2.41)	0.0018* (1.80)	0.0014 (1.31)
Constant	0.0598*** (3.76)	0.0320 (0.48)	-0.0220 (-0.33)	0.0027 (0.17)	-0.0313 (-0.78)	-0.0939* (-1.94)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	783	731	544	783	731	544
adj. R ²	0.258	0.098	0.042	0.247	0.101	0.105

Table XI: Bidder Characteristics and Bid Price Accuracy. This table reports results of regression analyses on the determinants of bid price accuracy. Panel A reports estimates of OLS regressions in which the dependent variable is the continuous measure of bid price *inaccuracy*, defined as the percentage difference between bid price and offer price. Panel B reports estimates of a Probit model in which the left-hand side dependent variable is a dummy variable indicating whether the bid price is above offer price or not. Independent variables capture a variety of bidder characteristics such as the size of the bid, the timing of the bid, the participation frequency of the bidder, the type of the bidder as well as the distance measures between bidder and IPO firm and between bidders. We include issue level fixed effects for all specifications. Variable definitions are given in Appendix A. Reported are coefficients and t-values calculated using robust standard errors with clustering on issues. ***, **, and * indicate 1%, 5%, and 10% statistical significance levels, respectively.

	Panel A: OLS regression		Panel B: Probit regression	
	Dependent variable: % difference between bid price and offer price		Dependent variable: 1 if bid price > offer price, otherwise 0	
	(1)	(2)	(3)	(4)
Distance	0.0259*** (3.58)		-0.0537*** (-3.55)	
Distance (alternative)		0.0270*** (4.19)		-0.0567*** (-4.22)
Largest bid	0.0006 (0.10)	0.0005 (0.070)	-0.001 (-0.07)	-0.0006 (-0.04)
Second largest bid	-0.0128 (-1.34)	-0.0129 (-1.35)	0.0271 (1.29)	0.0273 (1.31)
Early bid	0.0328*** (3.70)	0.0331*** (3.73)	-0.0572*** (-3.00)	-0.0577*** (-3.03)
Late bid	-0.0441*** (-4.82)	-0.0441*** (-4.82)	0.0849*** (4.42)	0.0849*** (4.43)
High frequency	-0.0783*** (-2.69)	-0.0782*** (-2.69)	0.1834*** (2.79)	0.1834*** (2.79)
Medium frequency	-0.0843*** (-3.21)	-0.0844*** (-3.22)	0.1891*** (3.18)	0.1894*** (3.18)
Type_IC	0.111*** (7.49)	0.112*** (7.50)	-0.2485*** (-7.87)	-0.2497*** (-7.89)
Type_FF	-0.0668*** (-6.57)	-0.0665*** (-6.54)	0.1181*** (5.67)	0.1175*** (5.64)
Type_ID	0.0648 (0.37)	0.0657 (0.38)	1.2220*** (15.48)	1.2220*** (15.52)
Type_FM	-0.0522*** (-5.29)	-0.0518*** (-5.25)	0.1042*** (4.98)	0.1033*** (4.93)
Type_QF	-0.0641* (-1.91)	-0.0632* (-1.89)	0.1037 (1.53)	0.1018 (1.50)
Type_II	-0.0639*** (-3.11)	-0.0629*** (-3.07)	0.1470*** (3.28)	0.1448*** (3.28)
Type_TC	-0.119*** (-10.28)	-0.119*** (-10.29)	0.2487*** (10.42)	0.2491*** (10.43)
Constant	0.704*** (12.68)	0.698*** (13.75)	-0.2608** (-2.22)	-0.2401** (-2.23)
IPO FE	Yes	Yes	Yes	Yes
N	123,819	123,819	123,819	123,819
adj. R2	0.102	0.102	0.091	0.091

Table XII: Dispersion and Other IPO Characteristics. This table reports regression results using *Turnover*, *Return Volatility*, *Oversubscription* in auction tranche and the *probability of IPO first-day closing price below its offer price* (denoted as *Offer-Break dummy*) as dependent variables, respectively. *Turnover* is defined as the proportion of first-day trading volume to the number of IPO shares offered. *Volatility* is calculated as the annualized 30-day volatility of market adjusted returns from the IPO listing date. *Oversubscription* is defined as the natural logarithm of the ratio of total bid quantities in auction tranche from institutional investors divided by the total number of shares offering in auction tranche. *Offer-Break dummy* is an indicator variable that takes on a value of one if the first-day closing price falls below the offer price and otherwise 0. The variable of interest is dispersion measure *STD*, defined as the ratio of the cross-sectional standard deviation of institutional investors' bidding price in the auction bidding process to the mean bidding price. Variable definitions are given in Appendix A. Reported are coefficients and *t*-values calculated using the industry clustered standard errors. ***, **, and * indicate 1%, 5%, and 10% statistical significance levels, respectively.

	OLS Regression			Probit
	Turnover	Volatility	Oversubscription	Offer-Break dummy
	(1)	(2)	(4)	(3)
STD	0.5043** (2.80)	0.2545** (2.30)	-2.7083*** (-6.89)	-5.3640*** (-7.35)
Firm age	0.0074 (1.59)	0.0038 (1.73)	0.0177* (1.91)	-0.1332*** (-3.16)
Offer size	-0.0977** (-5.97)	-0.0386** (-7.11)	-0.2697*** (-16.67)	0.6822*** (4.10)
Time gap	-0.0031 (-0.11)	0.0305*** (7.56)	-0.0086 (-0.31)	0.4677*** (2.86)
ROE	-0.0137 (-0.27)	0.0143*** (3.39)	-0.0163 (-0.24)	0.0499 (0.12)
market return	0.0073*** (28.11)	0.0037*** (9.27)	-0.0058** (-3.03)	-0.1014*** (-9.78)
Oversubscription	0.0459** (9.46)	0.0266*** (3.17)		-0.4708*** (-6.43)
Log # institutions	-0.0398** (-4.30)	0.0244*** (3.98)	1.6499*** (44.54)	0.1060 (1.00)
VC/PE dummy	-0.0093 (-0.76)	-0.0020 (-0.40)	0.0013 (0.03)	0.1604 (1.22)
ReguChange1	0.0878*** (3.89)	0.0946** (2.60)	-0.6851*** (-15.63)	-5.1771*** (-28.51)
ReguChange2	0.0373* (1.87)	-0.0081 (-0.74)	0.5105*** (9.40)	0.8663*** (4.16)
ChiNext dummy	-0.0172 (-1.68)	-0.0155** (-2.69)	0.0958*** (6.09)	0.1445** (2.49)
Lead reputation	0.0039 (1.19)	0.0016 (1.51)	-0.0163** (-2.72)	-0.0885*** (-2.75)
Constant	1.4062*** (18.29)	0.0292 (0.59)	-1.0947*** (-4.21)	-10.4677*** (-12.65)
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
<i>N</i>	783	783	783	766
adj. <i>R</i> ²	0.215	0.282	0.819	0.376